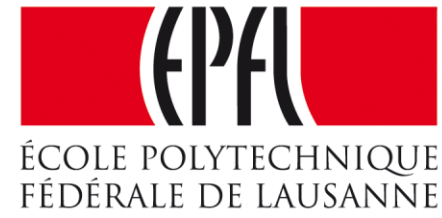


Week 4 – part 1 : Reduction of the Hodgkin-Huxley Model



Biological Modeling of Neural Networks

Week 4 – Reducing detail:

Two-dimensional neuron models

Wulfram Gerstner

EPFL, Lausanne, Switzerland

✓ 4.1 From Hodgkin-Huxley to 2D

- Overview: From 4 to 2 dimensions
- MathDetour 1: Exploiting similarities
- MathDetour 2: Separation of time scales

4.2 Phase Plane Analysis

- Role of nullclines

4.3 Analysis of a 2D Neuron Model

- constant input vs pulse input
- MathDetour 3: Stability of fixed points

4.4 Type I and II Neuron Models

Neuronal Dynamics – 4.2. Reduced Hodgkin-Huxley model

$$C \frac{du}{dt} = \underbrace{-g_{Na} m_0(u)^3 (1-w)(u - E_{Na})}_{I_{Na}} - \underbrace{g_K \left(\frac{w}{a}\right)^4 (u - E_K)}_{I_K} - \underbrace{g_l (u - E_l)}_{I_{leak}} + I(t)$$

$$\frac{dw}{dt} = -\frac{w - w_0(u)}{\tau_w(u)}$$

stimulus

$$\tau \frac{du}{dt} = F(u, w) + \overset{\text{stimulus}}{\downarrow} RI(t)$$

$$\tau_w \frac{dw}{dt} = G(u, w)$$

Neuronal Dynamics – 4.2. Nullclines of reduced HH model

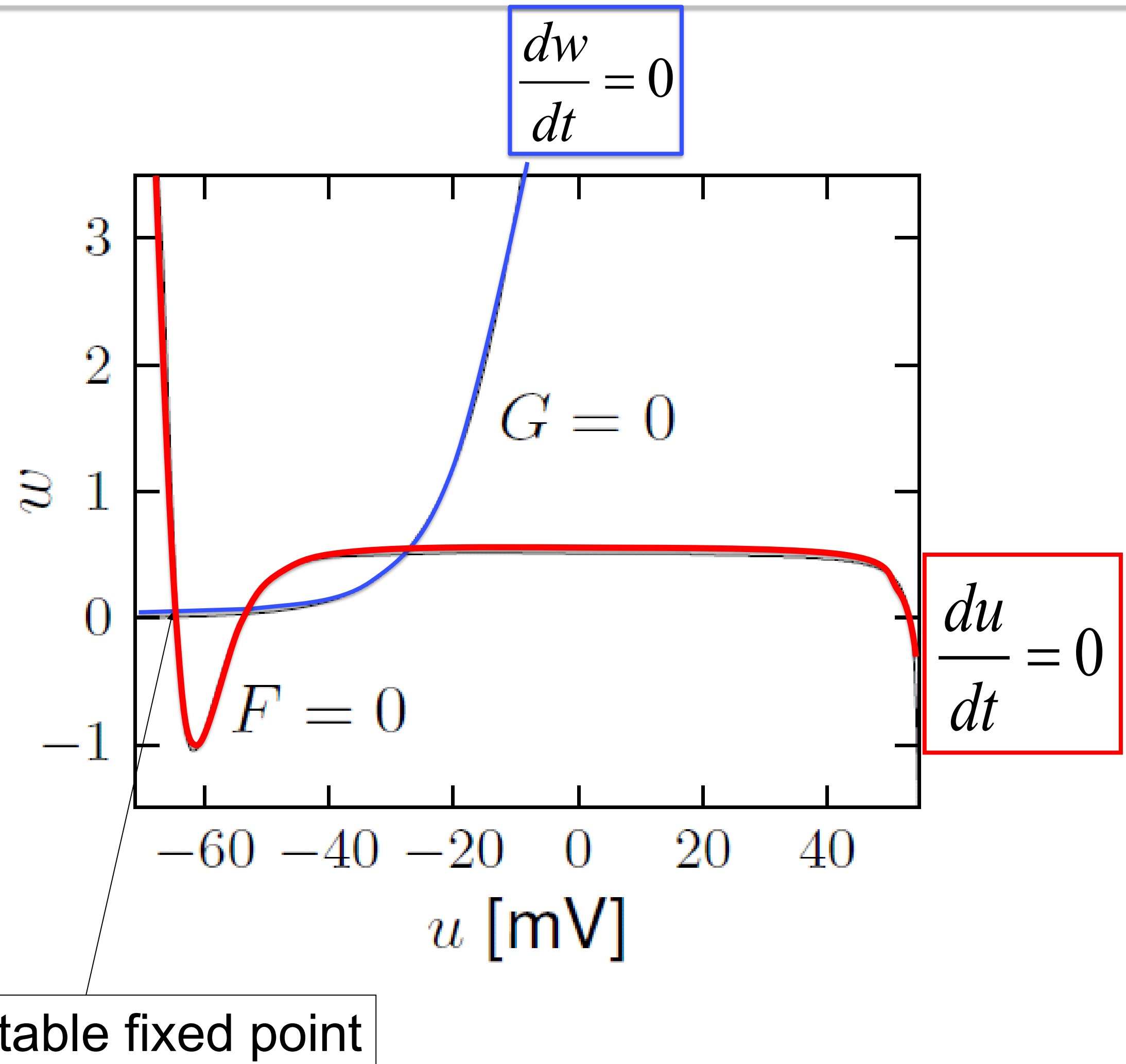
stimulus
↓

$$\tau \frac{du}{dt} = F(u, w) + RI(t)$$

$$\tau_w \frac{dw}{dt} = G(u, w)$$

u-nullcline $\frac{du}{dt} = 0$

w-nullcline $\frac{dw}{dt} = 0$



Neuronal Dynamics – 4.2. FitzHugh-Nagumo Model

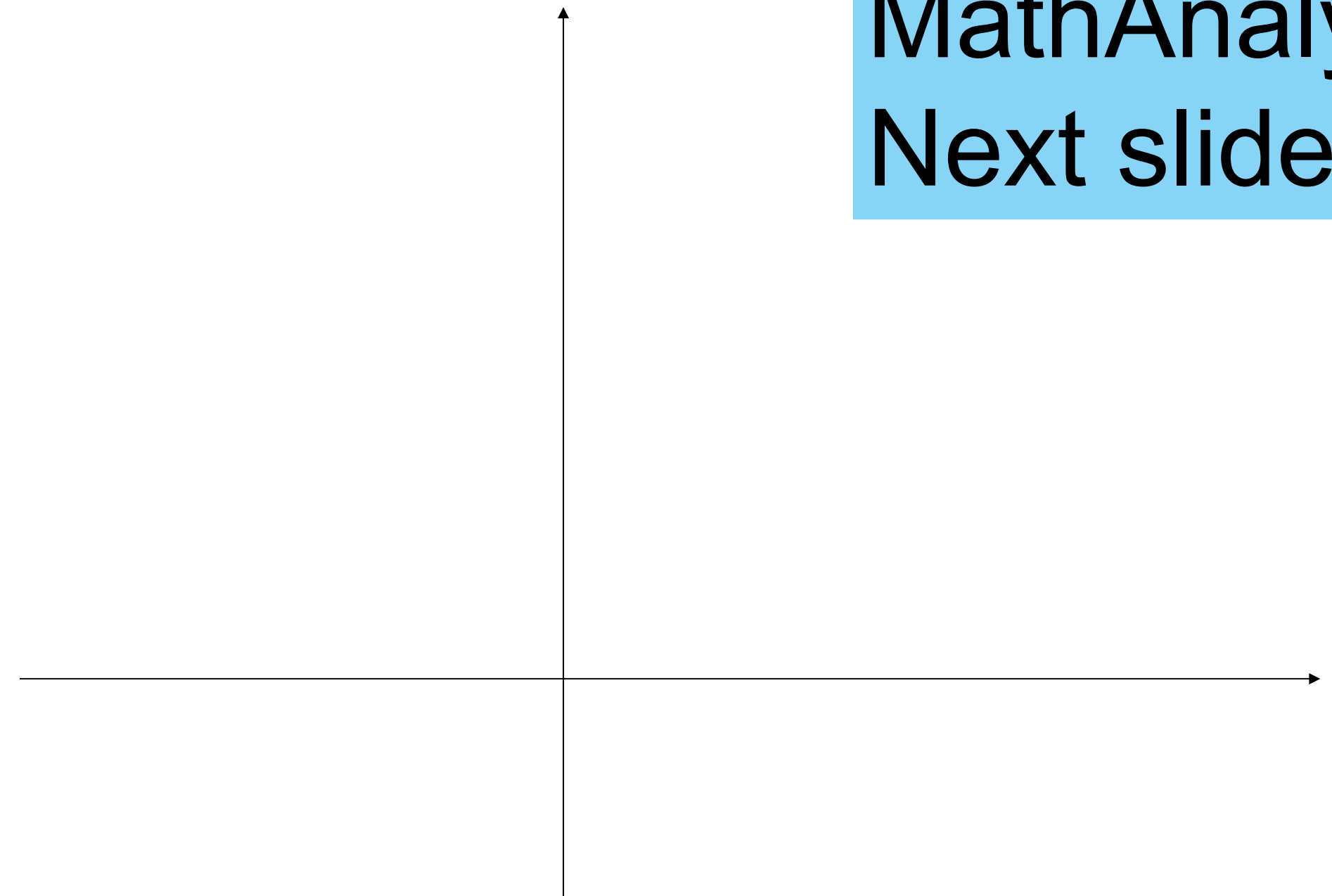
$$\tau \frac{du}{dt} = F(u, w) + RI(t)$$

$$= u - \frac{1}{3}u^3 - w + RI(t)$$

$$\tau_w \frac{dw}{dt} = G(u, w) = b_0 + b_1 u - w$$

u-nullcline

w-nullcline



MathAnalysis,
Next slide

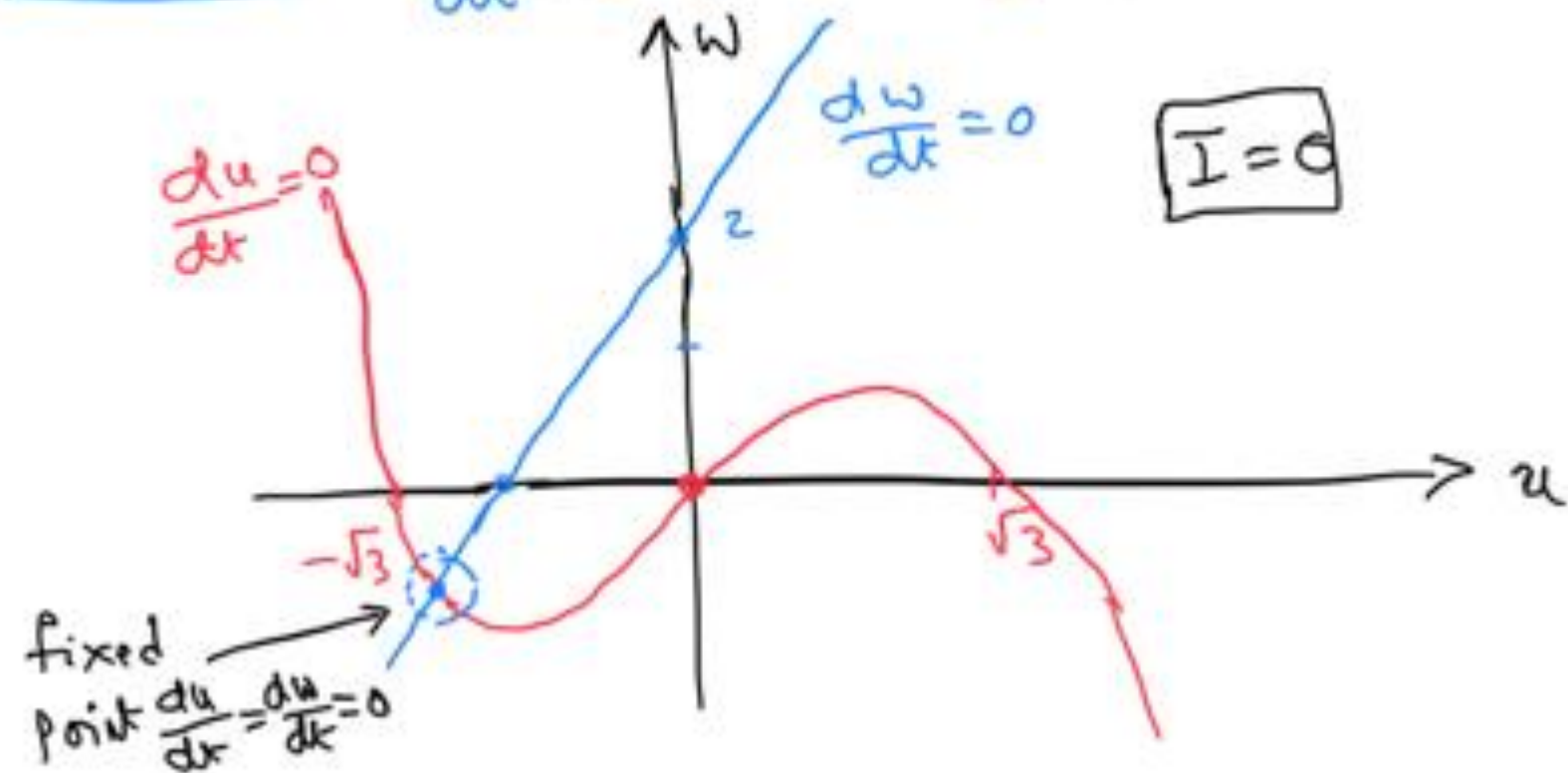
Phase-plane analysis of FHN model:

$$\begin{cases} \tau \frac{du}{dt} = F(u, w) + RI(t) \\ \tau_w \frac{dw}{dt} = G(u, w) = b_0 + b_1 u - w \end{cases} \Rightarrow \begin{matrix} I=0 \\ \text{indep of } I(t) \end{matrix}$$

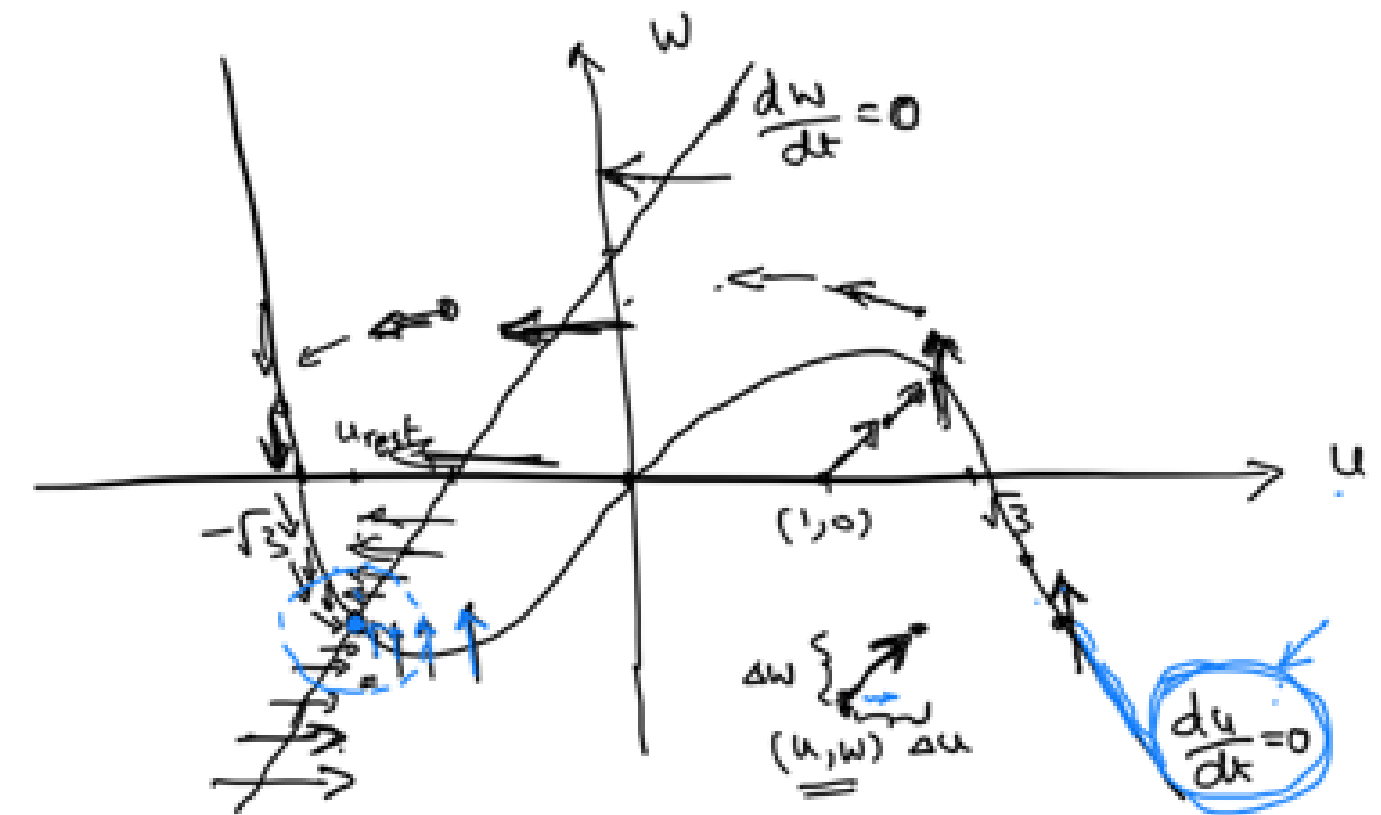
When $I=0$:

u-nullcline: $\frac{du}{dt} = 0 \Rightarrow w = u - \frac{1}{3}u^3 = u(1 - \frac{1}{3}u^2) \Rightarrow u=0 \text{ or } u=\pm\sqrt{3}$

w-nullcline: $\frac{dw}{dt} = 0 \Rightarrow w = b_0 + b_1 u$



Study the (flow) on this phase-plane:



For small timestep Δt : Flow at any point on u-w plane

$$\begin{pmatrix} \Delta u \\ \Delta w \end{pmatrix} = \begin{pmatrix} \frac{du}{dt} \cdot \Delta t \\ \frac{dw}{dt} \cdot \Delta t \end{pmatrix} = \begin{pmatrix} \frac{du}{dt} \\ \frac{dw}{dt} \end{pmatrix} \cdot \Delta t \quad \checkmark$$

Flow on nullcline:

u-nullcline: $\frac{du}{dt} = 0$

$$\begin{pmatrix} \Delta u \\ \Delta w \end{pmatrix} = \begin{pmatrix} 0 \\ \frac{dw}{dt} \end{pmatrix} \cdot \Delta t \quad \uparrow \downarrow$$

w-nullcline: $\frac{dw}{dt} = 0$

$$\begin{pmatrix} \Delta u \\ \Delta w \end{pmatrix} = \begin{pmatrix} \frac{du}{dt} \\ 0 \end{pmatrix} \cdot \Delta t \quad \leftarrow \rightarrow$$

Neuronal Dynamics – 4.2. flow arrows

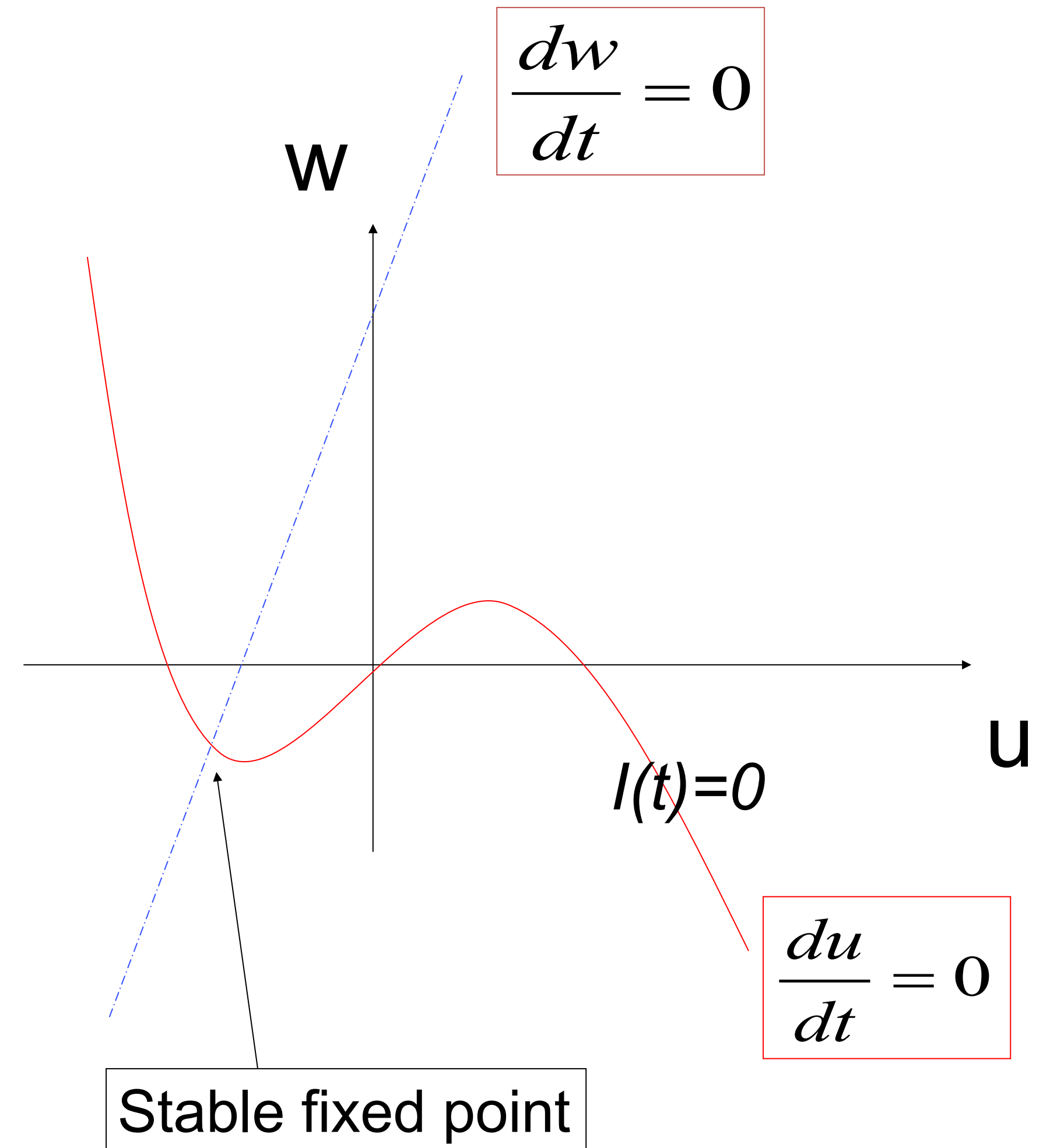
$$\tau \frac{du}{dt} = F(u, w) + RI(t) \quad \text{Stimulus } I=0$$

$$\tau_w \frac{dw}{dt} = G(u, w)$$

Consider change in small time step

Flow on nullcline

Flow in regions between nullclines

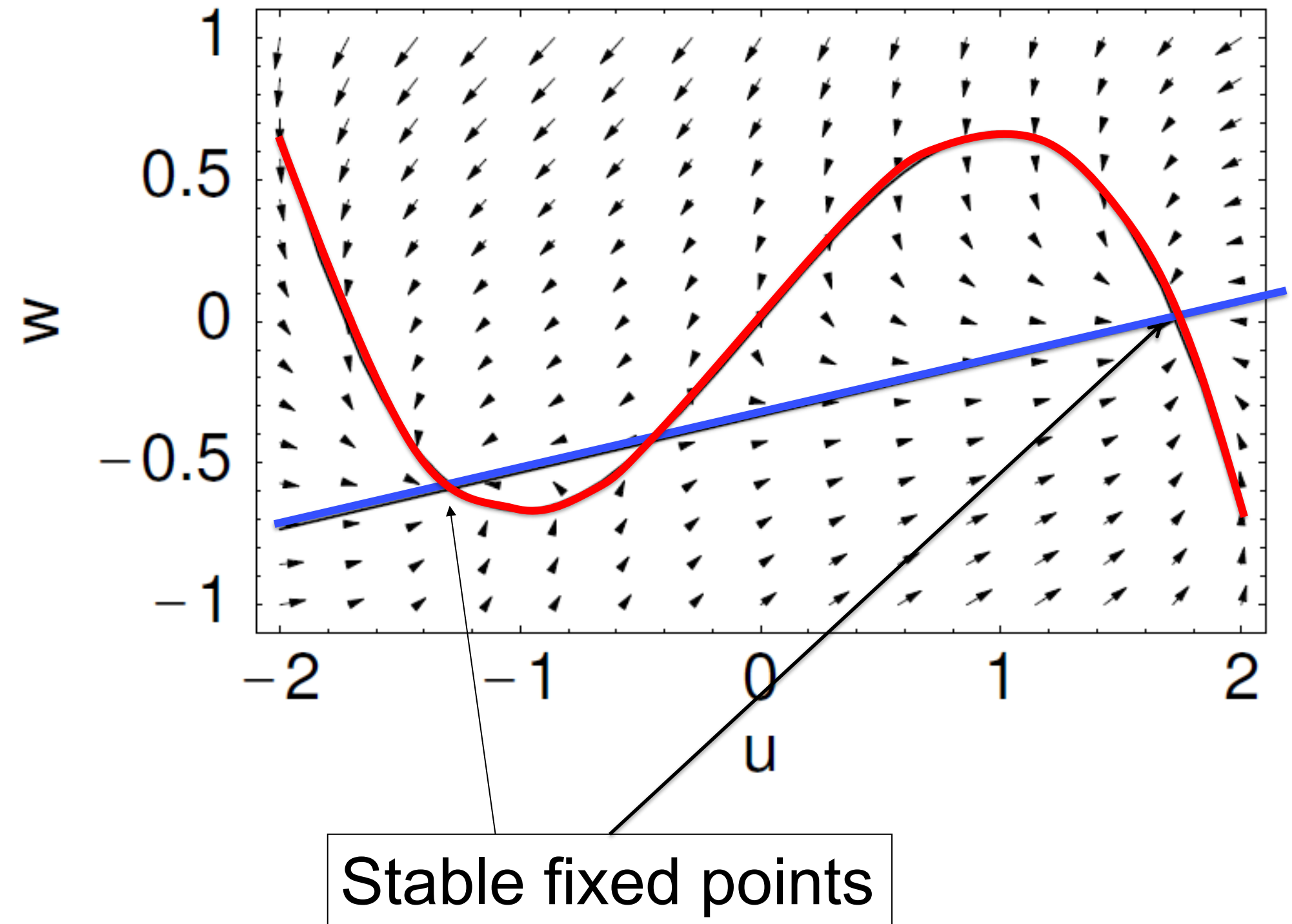


Neuronal Dynamics – 4.2. flow arrows

$$\tau \frac{du}{dt} = F(u, w) + RI(t)$$

$$= u - \frac{1}{3}u^3 - w + RI(t)$$

$$\tau_w \frac{dw}{dt} = G(u, w) = b_0 + b_1 u - w$$



Neuronal Dynamics – 4.2. Phase Plane Analysis/nullclines

2-dimensional equation ^{stimulus}

$$\tau \frac{du}{dt} = F(u, w) + RI(t)$$

$$\tau_w \frac{dw}{dt} = G(u, w)$$

Enables graphical analysis!

Important role of

- nullclines
- flow arrows

Class Participation Questions

Choose Correct answer(s). multiple correct choices possible!

A. u-Nullclines

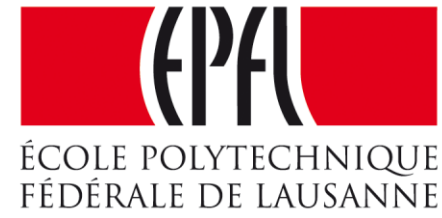
- ☐ On the u-nullcline, arrows are always vertical
- ☐ On the u-nullcline, arrows point always vertically upward
- ☐ On the u-nullcline, arrows are always horizontal
- ☐ On the u-nullcline, arrows point always to the left
- ☐ On the u-nullcline, arrows point always to the right

B. w-Nullclines

- ☐ On the w-nullcline, arrows are always vertical
- ☐ On the w-nullcline, arrows point always vertically upward
- ☐ On the w-nullcline, arrows are always horizontal
- ☐ On the w-nullcline, arrows point always to the left
- ☐ On the w-nullcline, arrows point always to the right
- ☐ On the w-nullcline, arrows can point in an arbitrary direction

Pl submit on moodle within 45 mins after today's class

Week 4 – part 1 : Reduction of the Hodgkin-Huxley Model



Biological Modeling of Neural Networks

Week 4 – Reducing detail:

Two-dimensional neuron models

Wulfram Gerstner

EPFL, Lausanne, Switzerland

✓ 4.1 From Hodgkin-Huxley to 2D

- Overview: From 4 to 2 dimensions
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✓ 4.2 Phase Plane Analysis

- Role of nullclines

4.3 Analysis of a 2D Neuron Model

- constant input vs pulse input
- MathDetour 3: Stability of fixed points

4.4 Type I and II Neuron Models Part 2!

Neuronal Dynamics – 4.3. Analysis of a 2D neuron model

2-dimensional equation ^{stimulus}

$$\tau \frac{du}{dt} = F(u, w) + RI(t)$$

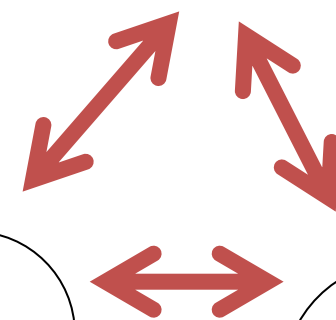
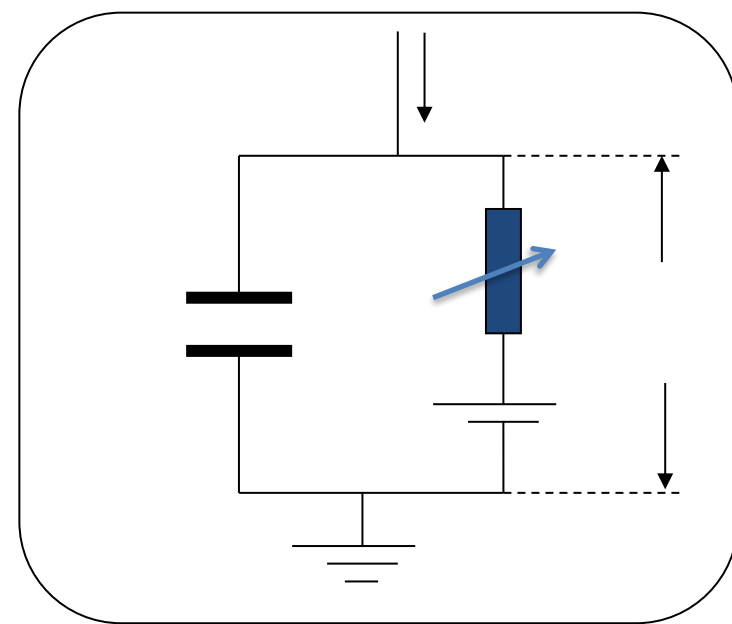
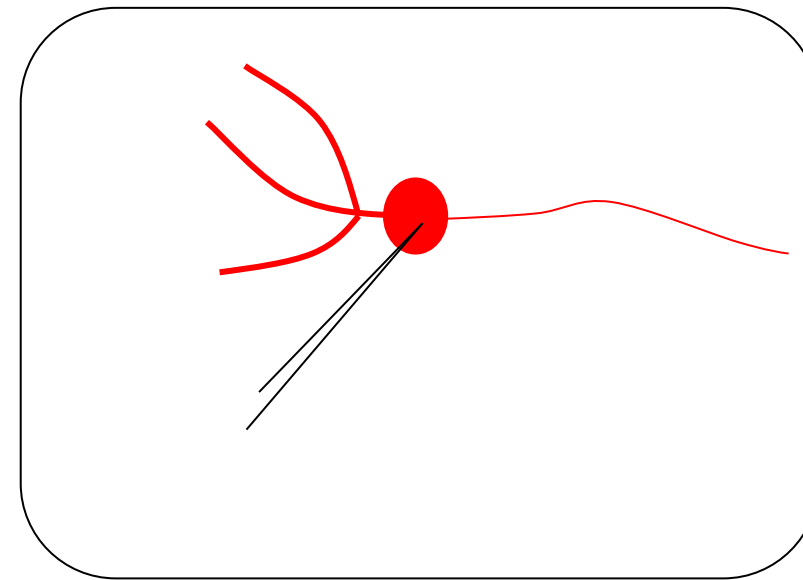
$$\tau_w \frac{dw}{dt} = G(u, w)$$

2 important input scenarios

- Pulse input
- Constant input

Enables graphical analysis!

Neuronal Dynamics – 4.3. 2D neuron model : Pulse input



$$\tau \cdot \frac{d}{dt} u = F(u, w) + RI$$
$$\tau_w \frac{d}{dt} w = G(u, w)$$

pulse input

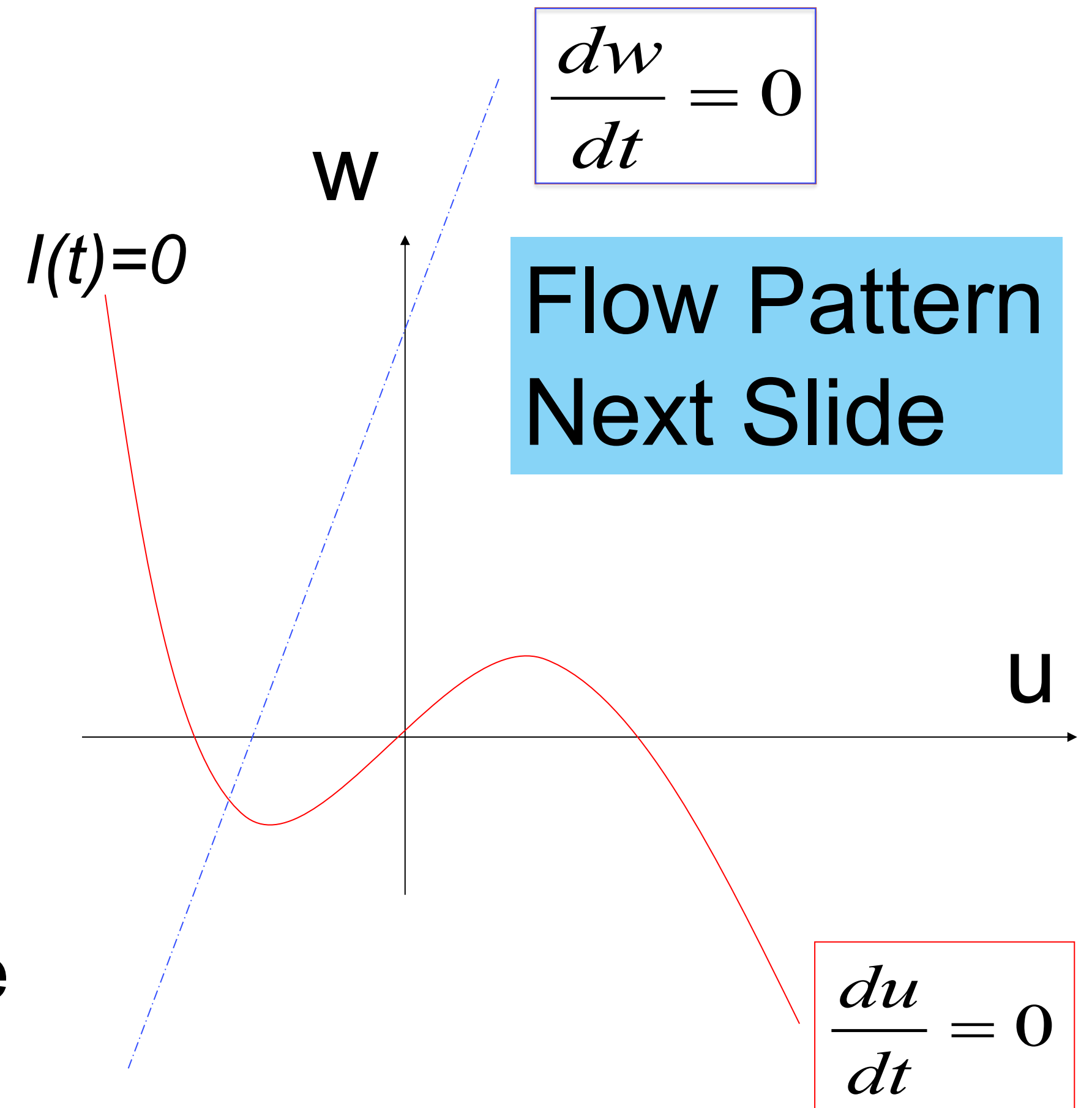


Neuronal Dynamics – 4.3. FitzHugh-Nagumo Model : Pulse input

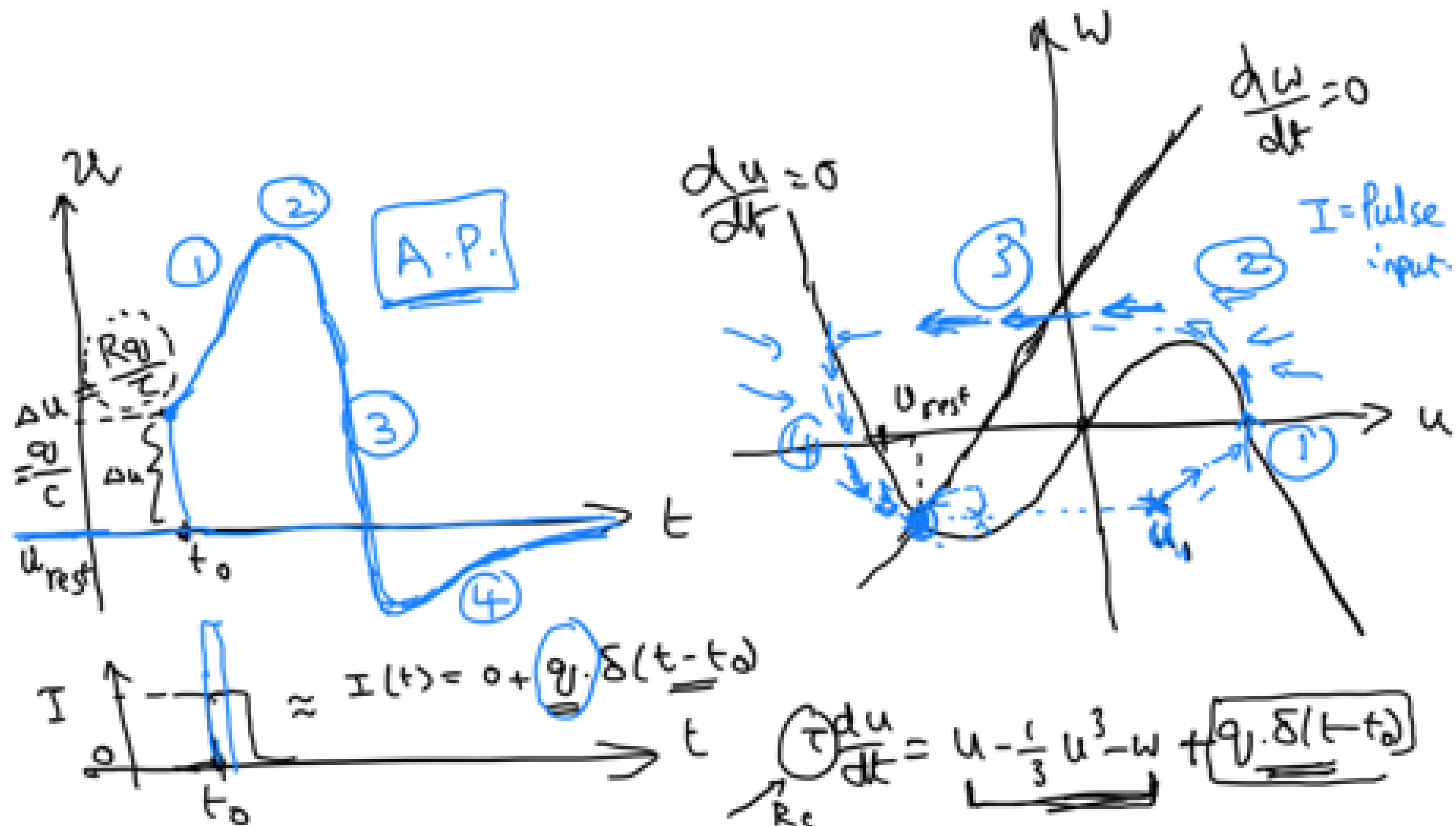
$$\tau \frac{du}{dt} = F(u, w) + RI(t) = u - \frac{1}{3}u^3 - w + RI(t)$$

$$\tau_w \frac{dw}{dt} = G(u, w) = b_0 + b_1 u - w$$

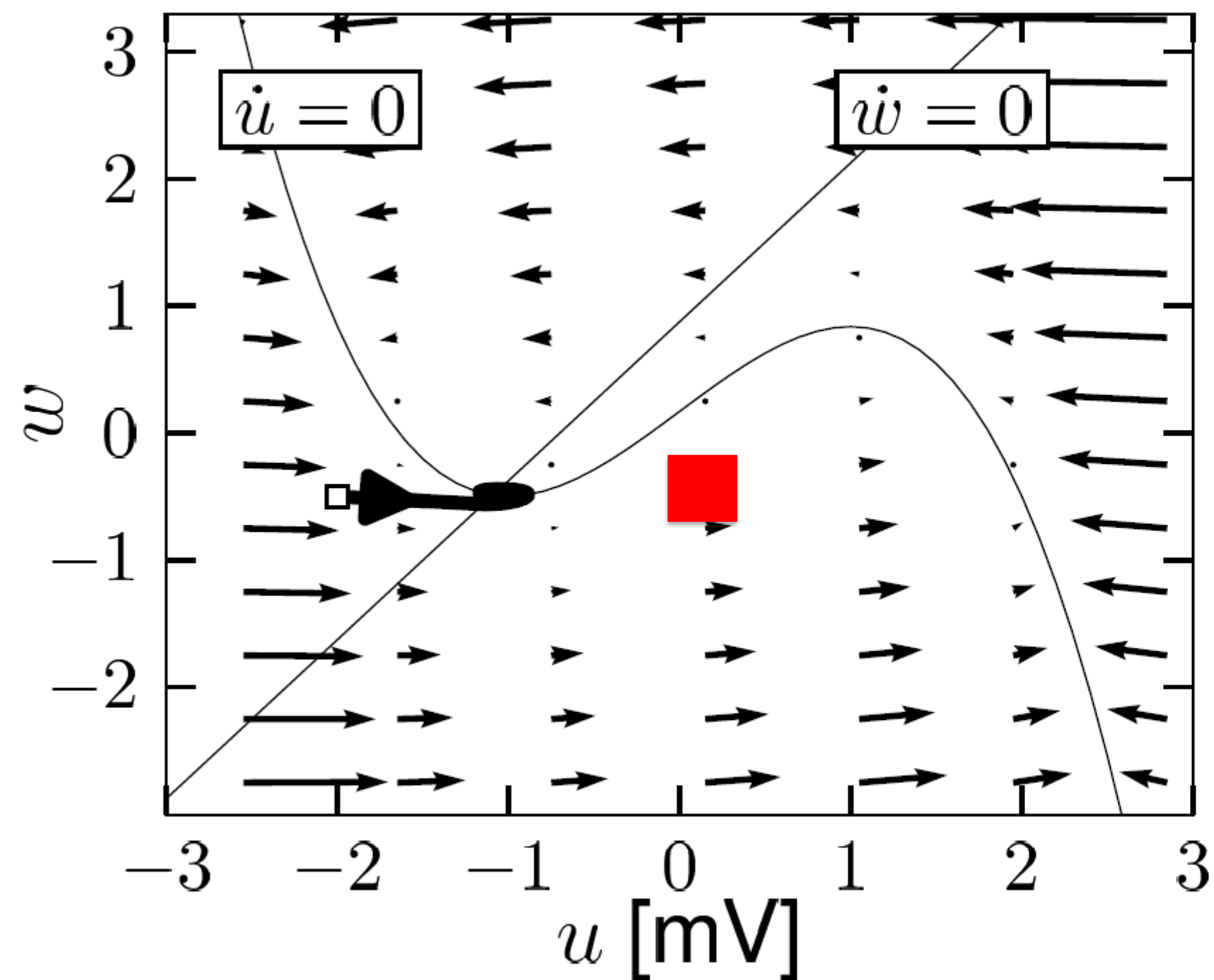
pulse input $I(t)$ Pulse input: jump of voltage



Pulse Input: Flow pattern on Phase-plane:



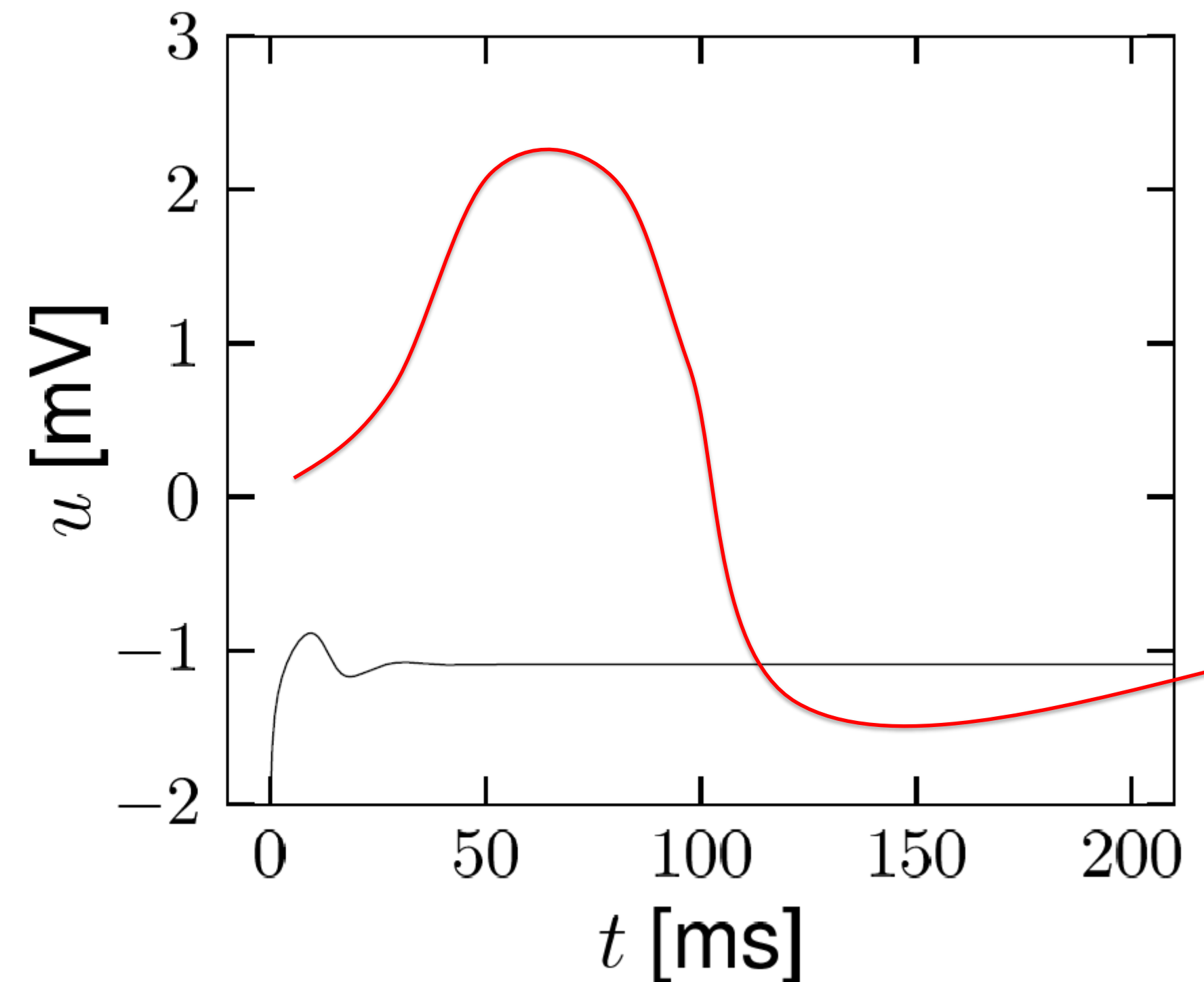
Neuronal Dynamics – 4.3. FitzHugh-Nagumo Model : Pulse input



FN model with $b_0 = 0.9; b_1 = 1.0$

Pulse input: jump of voltage/initial condition

B



*Image: Neuronal Dynamics,
Gerstner et al.,
Cambridge Univ. Press (2014)*

Neuronal Dynamics – 4.3. FitzHugh-Nagumo Model

2 important input scenarios

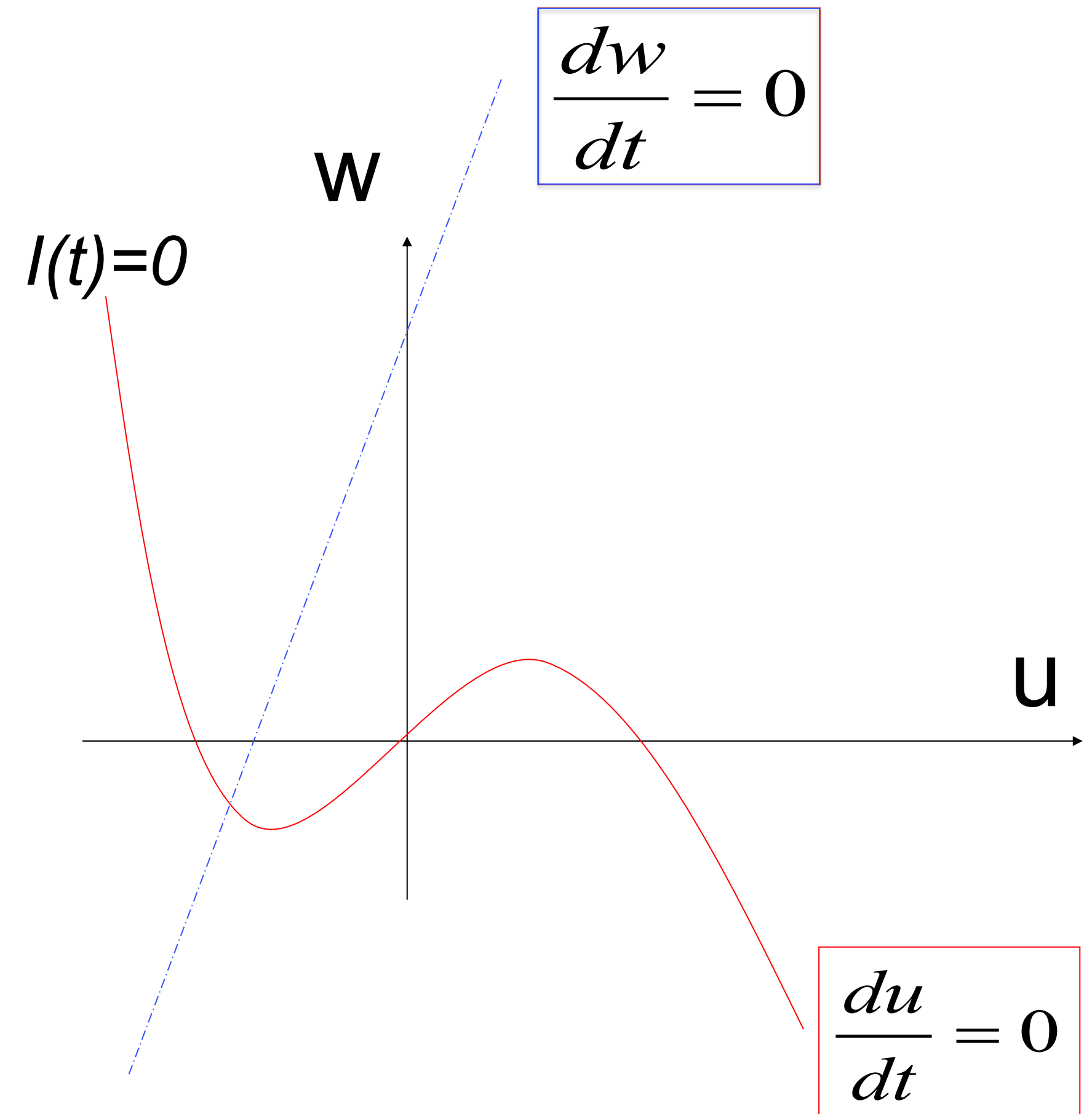
Pulse input: **DONE!**

- jump of voltage
- 'new initial condition'
- spike generation for large input pulses

constant input:

- graphics?
- spikes?
- repetitive firing?

Now



Neuronal Dynamics – 4.3. FitzHugh-Nagumo Model: Constant input

$$\tau \frac{du}{dt} = F(u, w) + RI_0$$

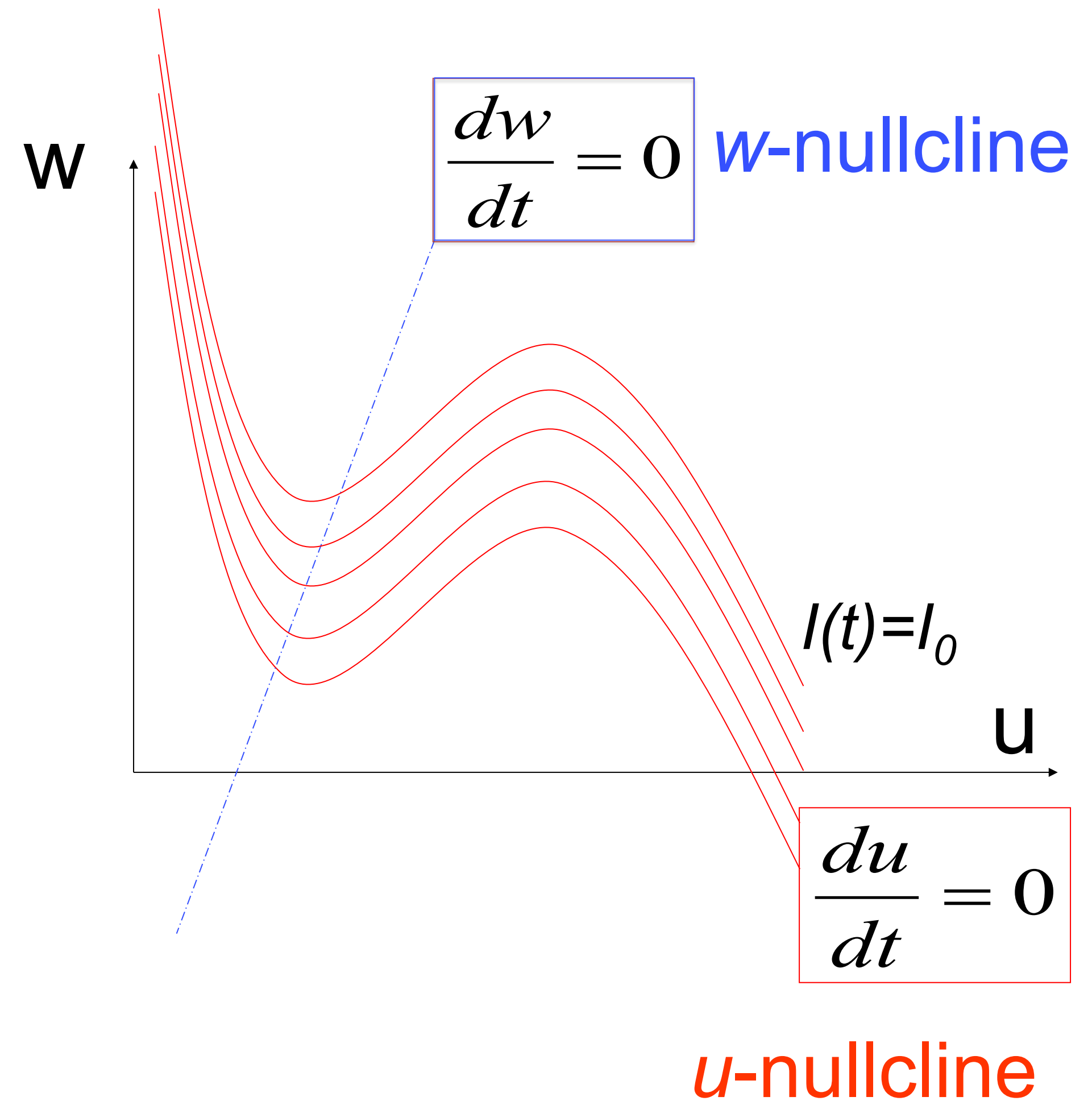
$$= u - \frac{1}{3}u^3 - w + RI_0$$

$$\tau_w \frac{dw}{dt} = G(u, w) = b_0 + b_1 u - w$$

Intersection point (fixed point)

-moves

-changes Stability



Neuronal Dynamics – 4.3. FitzHugh-Nagumo Model: Constant input

$$\tau \frac{du}{dt} = F(u, w) + RI_0$$

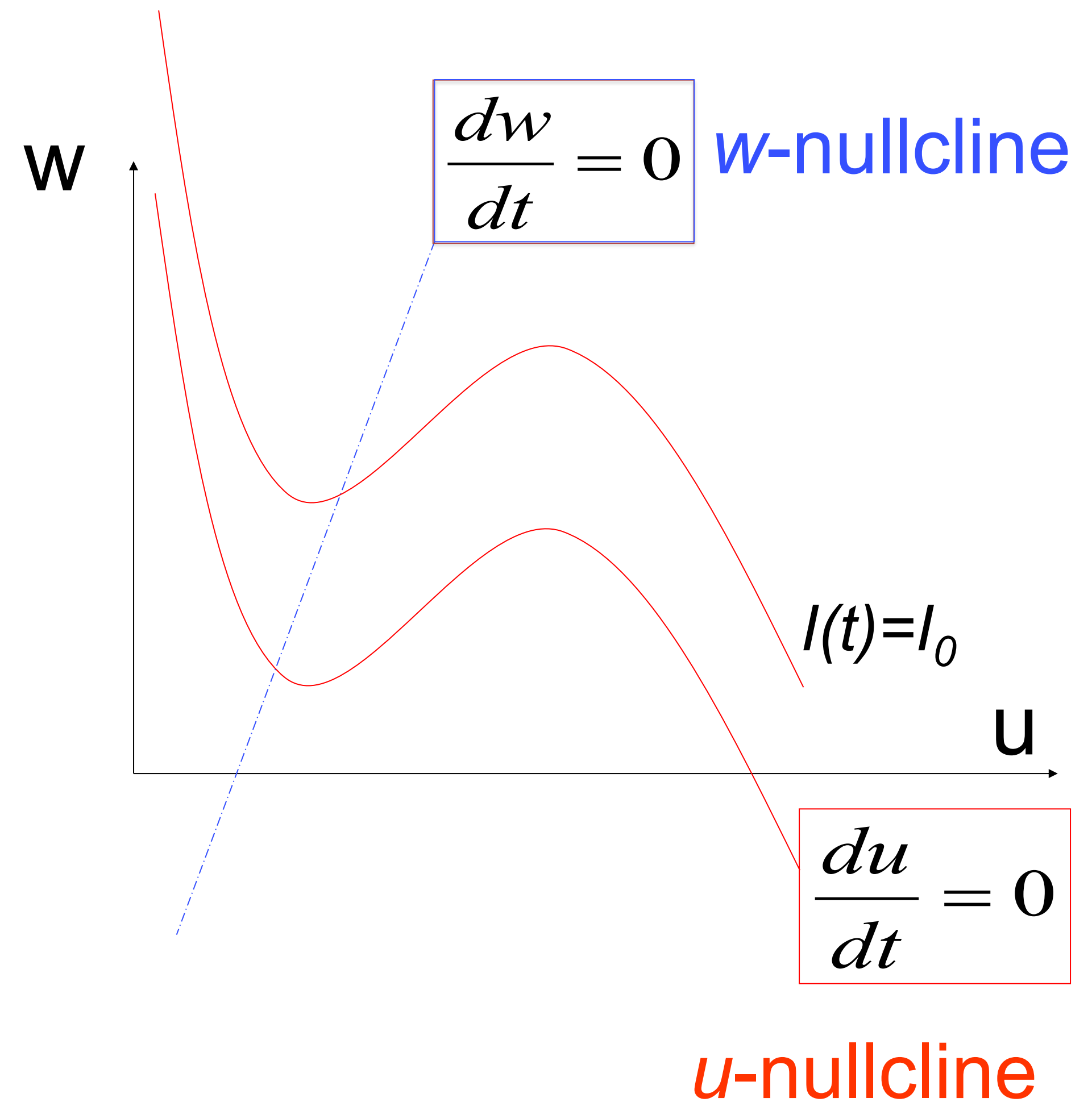
$$= u - \frac{1}{3}u^3 - w + RI_0$$

$$\tau_w \frac{dw}{dt} = G(u, w) = b_0 + b_1 u - w$$

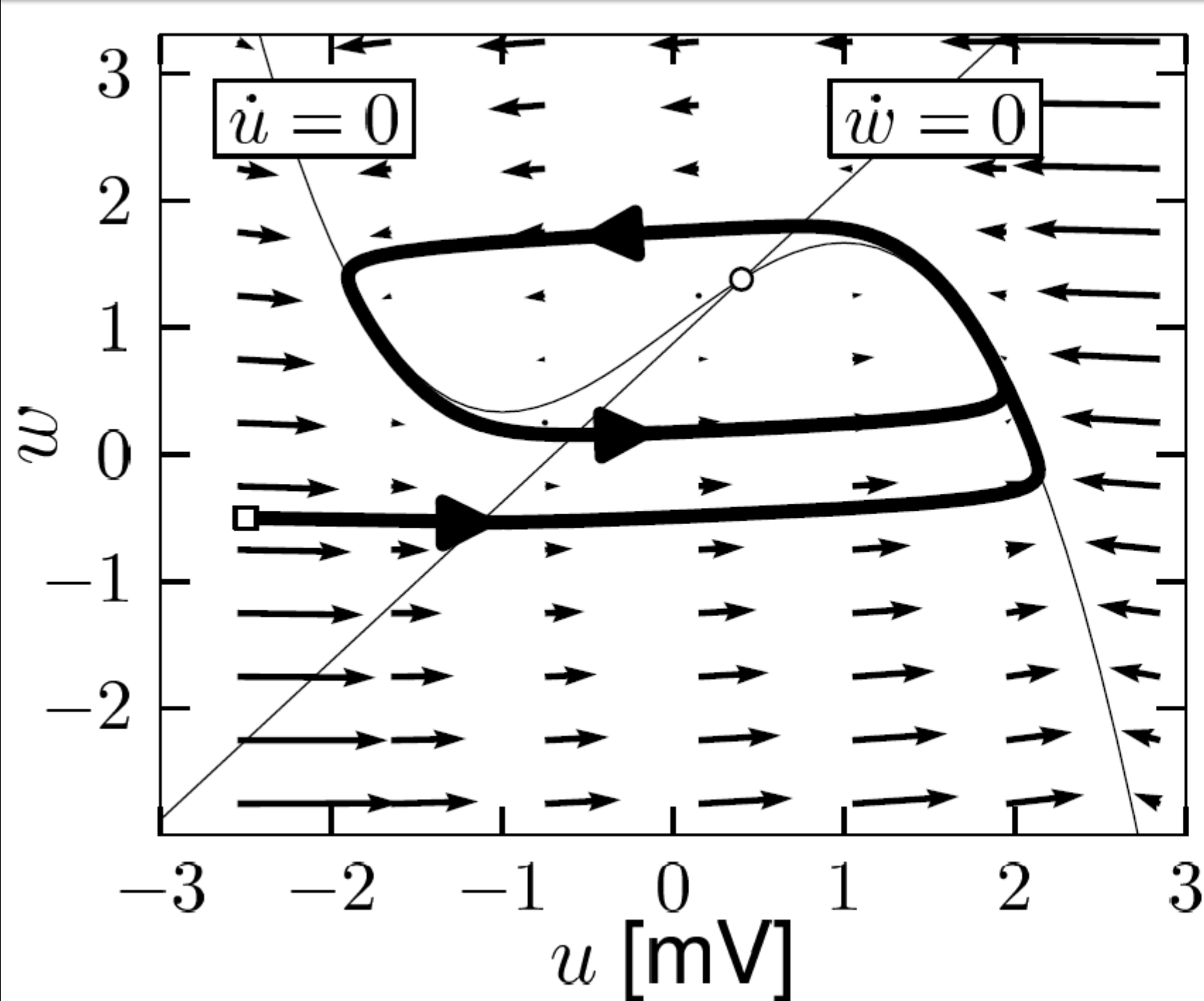
Intersection point (fixed point)

-moves

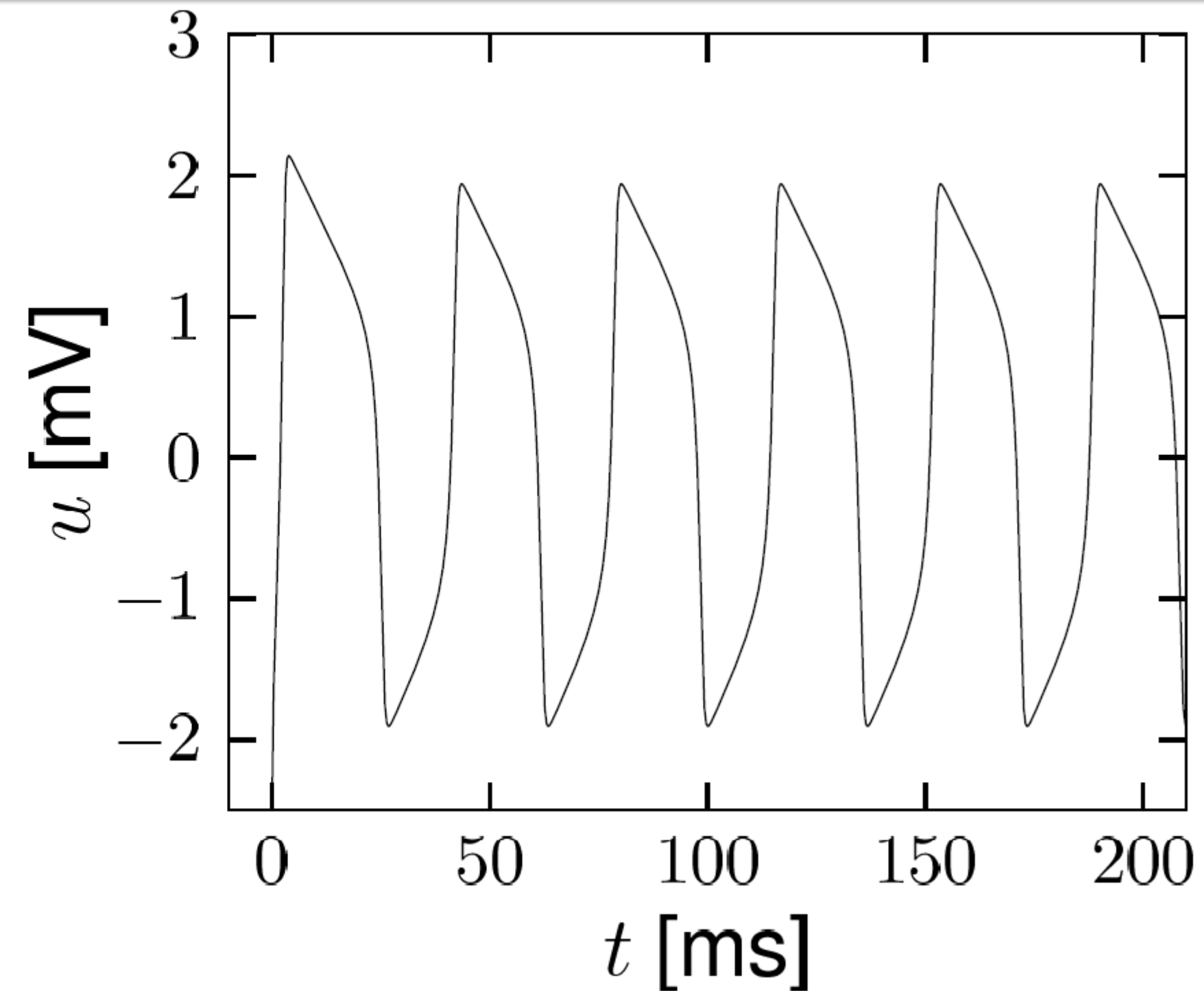
-changes Stability



Neuronal Dynamics – 4.3. FitzHugh-Nagumo Model : Constant input



D



FN model with $b_0 = 0.9; b_1 = 1.0; RI_0 = 2$
constant input: u -nullcline moves
limit cycle

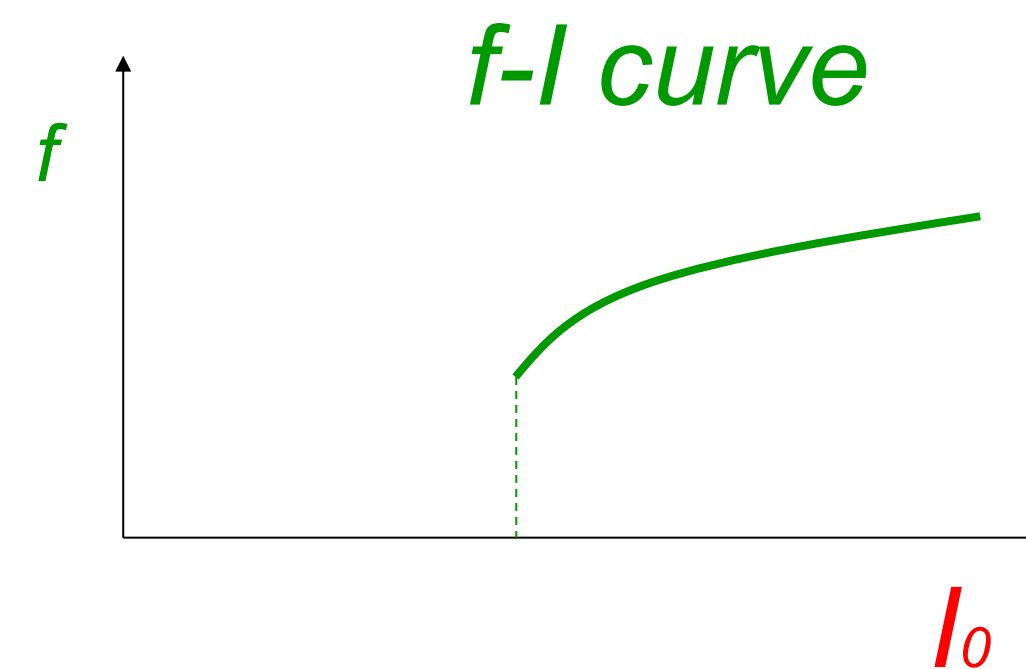
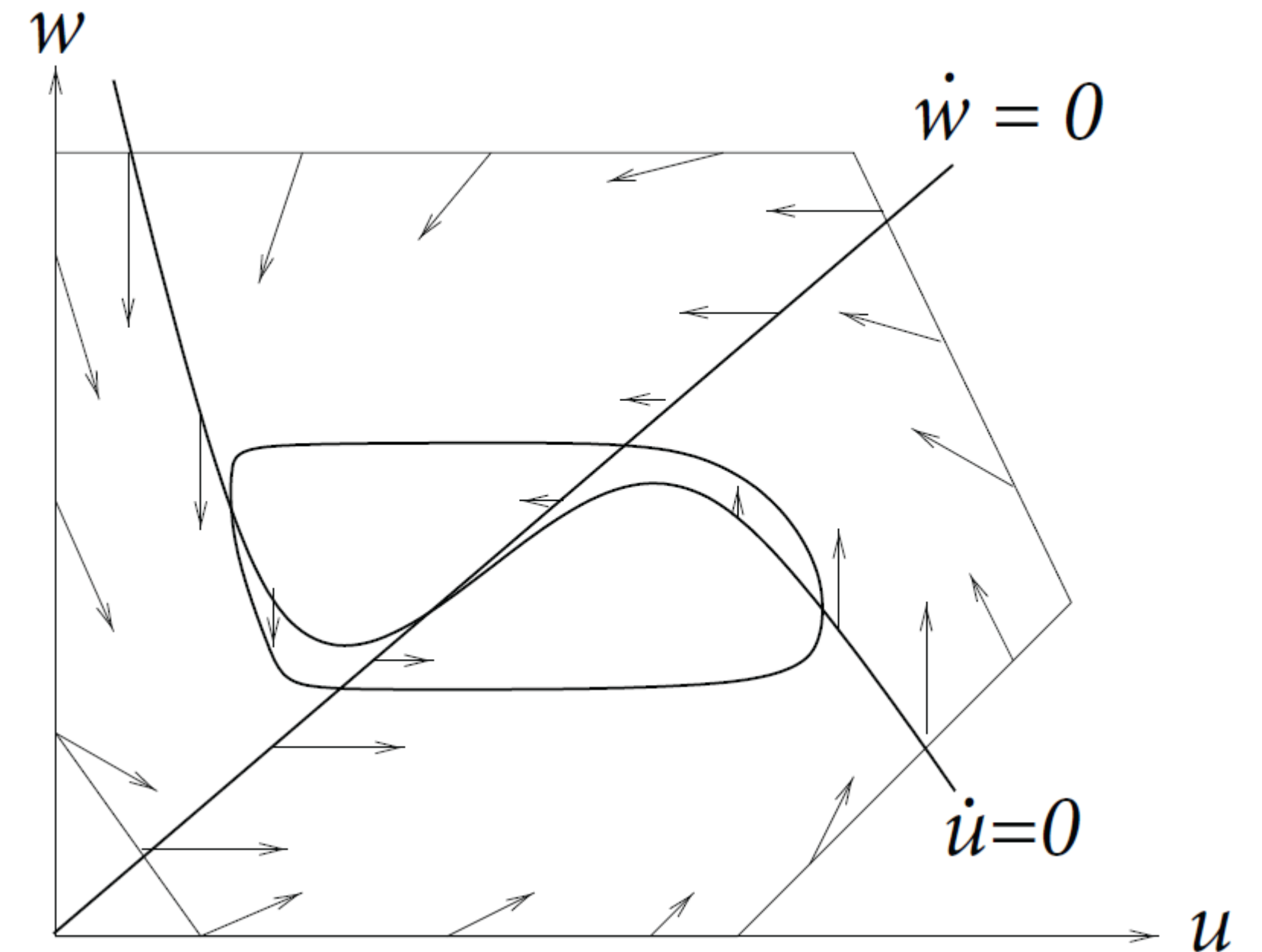


Image:
Neuronal Dynamics,
Gerstner et al.,
CUP (2014)

Neuronal Dynamics – 4.3. Limit Cycle

**In 2-dimensional equations,
a limit cycle must exist, if we can
find a surface**

- containing one unstable fixed point
- no other fixed point
- bounding box with inward flow
→ limit cycle (*Poincare Bendixson*)



*Image: Neuronal Dynamics,
Gerstner et al.,
Cambridge Univ. Press (2014)*

Neuronal Dynamics – 4.3. Analysis of a 2D neuron model

2-dimensional equation ^{stimulus}

$$\tau \frac{du}{dt} = F(u, w) + RI(t)$$

$$\tau_w \frac{dw}{dt} = G(u, w)$$

2 important input scenarios

- Pulse input
 - AP firing (or not)
- Constant input
 - Repetitive firing (or not)

Enables graphical analysis!

Neuronal Dynamics – Quiz 4.4.

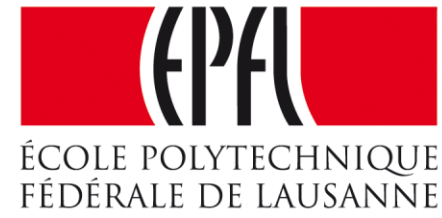
A. Short current pulses. In a 2-dimensional neuron model, the effect of a delta current pulse can be analyzed

- ☐ By moving the u-nullcline vertically upward
- ☐ By moving the w-nullcline vertically upward
- ☐ As a potential change in the stability or number of the fixed point(s)
- ☐ As a new initial condition
- ☐ By following the flow of arrows in the appropriate phase plane diagram

B. Constant current. In a 2-dimensional neuron model, the effect of a constant current can be analyzed

- ☐ By moving the u-nullcline vertically upward
- ☐ By moving the w-nullcline vertically upward
- ☐ As a potential change in the stability or number of the fixed point(s)
- ☐ By following the flow of arrows in the appropriate phase plane diagram

Week 4 – part 1 : Reduction of the Hodgkin-Huxley Model



Biological Modeling of Neural Networks

Week 4 – Reducing detail:

Two-dimensional neuron models

Wulfram Gerstner

EPFL, Lausanne, Switzerland

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✓ 4.2 Phase Plane Analysis

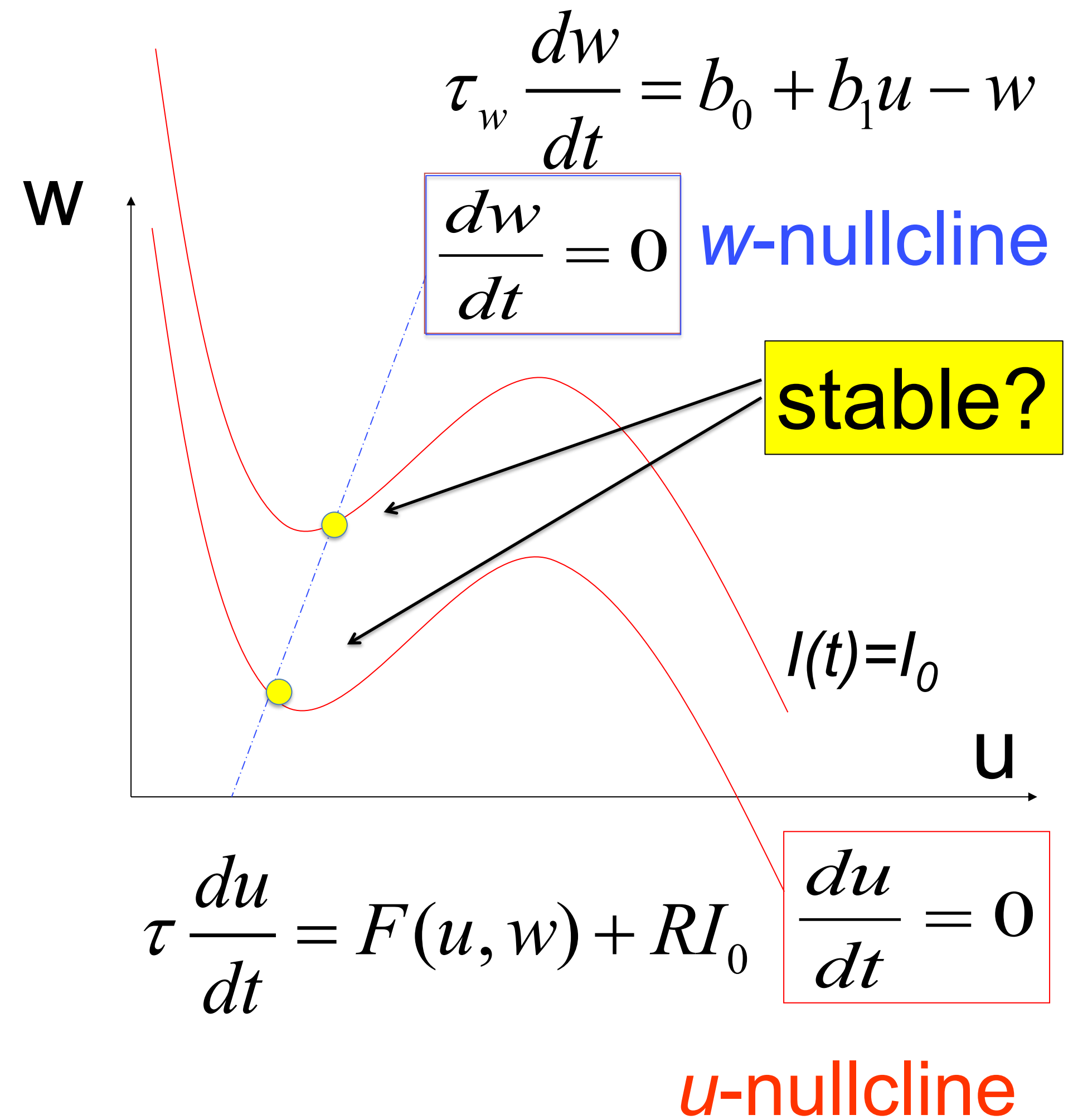
- Role of nullclines

4.3 Analysis of a 2D Neuron Model

- ✓ - constant input vs pulse input
- MathDetour 3: Stability of fixed points

4.4 Type I and II Neuron Models

Neuronal Dynamics – Detour 4.3 : Stability of fixed points.



Neuronal Dynamics – 4.3 Detour. Stability of fixed points

2-dimensional equation

$$\tau \frac{du}{dt} = F(u, w) + RI_0$$

$$\tau_w \frac{dw}{dt} = G(u, w)$$

How to determine stability
of fixed point?

Neuronal Dynamics – 4.3 Detour. Stability of fixed points

$$\tau \frac{du}{dt} = au - w + I_0$$

$$\tau_w \frac{dw}{dt} = cu - w$$

stimulus



W



$$\frac{dw}{dt} = 0$$

$$\frac{du}{dt} = 0$$

$l(t) = I_0$

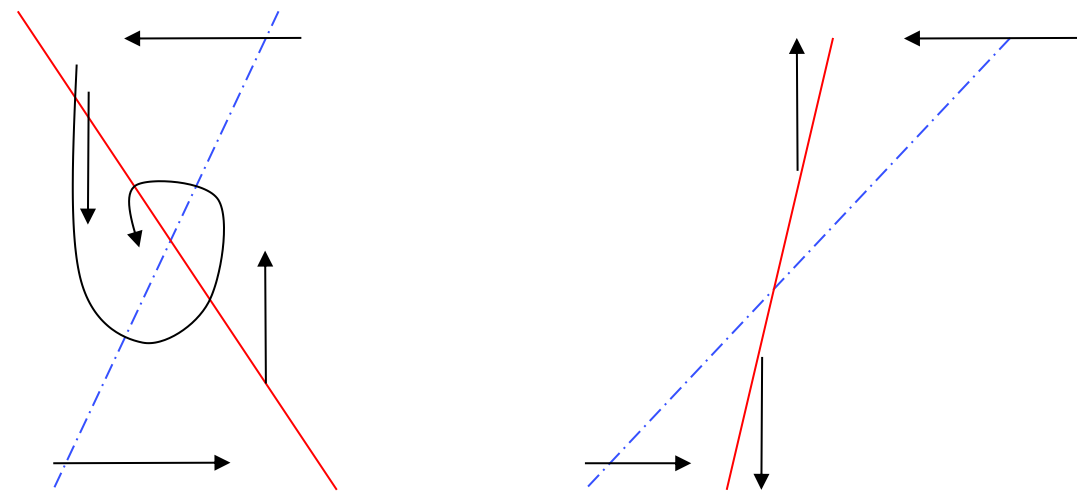
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Neuronal Dynamics – 4.3 Detour. Stability of fixed points

$$\tau \frac{du}{dt} = F(u, w) + RI_0$$

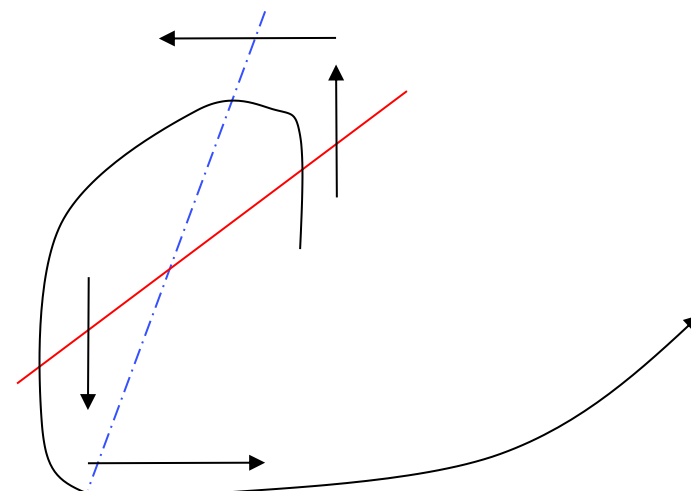
$$\tau_w \frac{dw}{dt} = G(u, w)$$

zoom in:

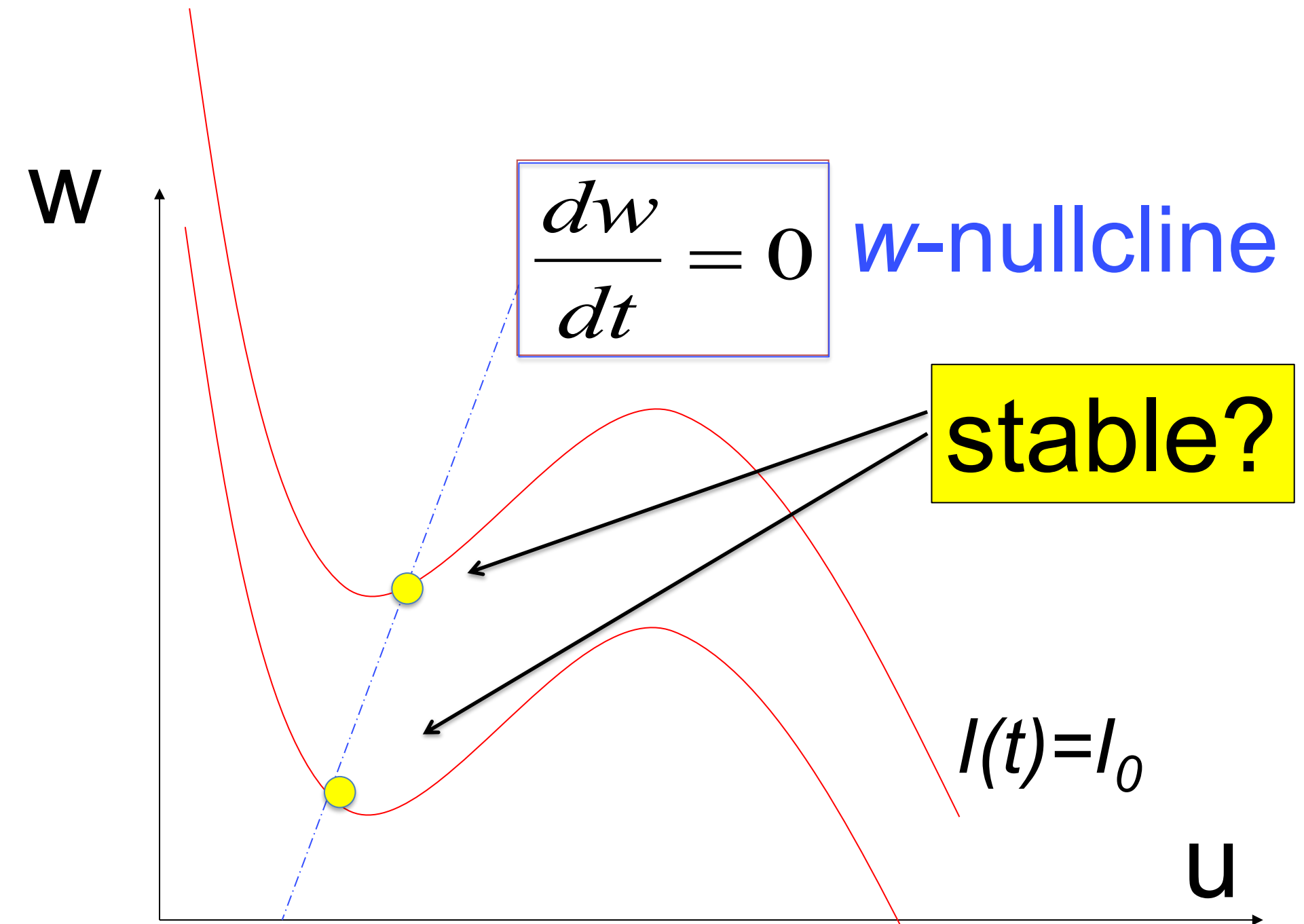


stable

saddle



unstable



Math derivation
now

u-nullcline

Neuronal Dynamics – 4.3 Detour. Stability of fixed points

$$\tau \frac{du}{dt} = F(u, w) + RI_0$$

$$\tau_w \frac{dw}{dt} = G(u, w)$$

zoom in:

$$x = u - u_0$$

$$y = w - w_0$$

Fixed point at (u_0, w_0)

At fixed point

$$0 = F(u_0, w_0) + RI_0$$

$$0 = G(u_0, w_0)$$

Stability Analysis of Fixed Points:

① $\tau \frac{du}{dt} = F(\underline{u}, \underline{w}) + RI_0$ let's assume fixed point is at (u_0, w_0)

② $\tau_w \frac{dw}{dt} = G(\underline{u}, \underline{w})$ @ fixed point:

Take a small neighbourhood around the fixed point: (u_0, w_0)

$$\begin{cases} \frac{du}{dt} = 0 \Rightarrow F(u, w) + RI_0 = 0 \\ \frac{dw}{dt} = 0 \Rightarrow G(u, w) = 0 \end{cases}$$

$$\begin{aligned} x &= u - u_0 \\ y &= w - w_0 \end{aligned} \quad \left| \begin{array}{l} u = x + u_0 \\ w = y + w_0 \end{array} \right.$$

$$\begin{aligned} \frac{dx}{dt} &= \frac{d(u - u_0)}{dt} = \frac{du}{dt} \\ \frac{dy}{dt} &= \frac{d(w - w_0)}{dt} = \frac{dw}{dt} \end{aligned}$$

$$\begin{aligned} \tau \frac{du}{dt} &= F(\underline{u}, \underline{w}) + RI_0 \\ &= F(u_0 + x, w_0 + y) + RI_0 \end{aligned}$$

Taylor Series expansion, in the small n.hood at (u_0, w_0) , can ignore H.O. terms

$$\begin{aligned} &= F(u_0, w_0) + \frac{\partial F}{\partial u} \bigg|_{(u_0, w_0)} \cdot \frac{x}{1!} + \frac{\partial F}{\partial w} \bigg|_{(u_0, w_0)} \cdot \frac{y}{1!} + \text{H.O. terms} + RI_0 \\ &= \underbrace{F(u_0, w_0) + RI_0}_0 + \frac{\partial F}{\partial u} \bigg|_{(u_0, w_0)} \cdot x + \frac{\partial F}{\partial w} \bigg|_{(u_0, w_0)} \cdot y = F_u \cdot x + F_w \cdot y \end{aligned}$$

Taylor series expansion of (2) $\tau_w \frac{dw}{dt} = G(u, w)$

$$\hookrightarrow \tau_w \frac{dw}{dt} = G_u x + G_w y \quad \frac{\partial G}{\partial u} \Big|_{(u_0, w_0)} \quad \frac{\partial G}{\partial w} \Big|_{(u_0, w_0)}$$

Vector form:

Let $\vec{x} = \begin{pmatrix} x \\ y \end{pmatrix}$

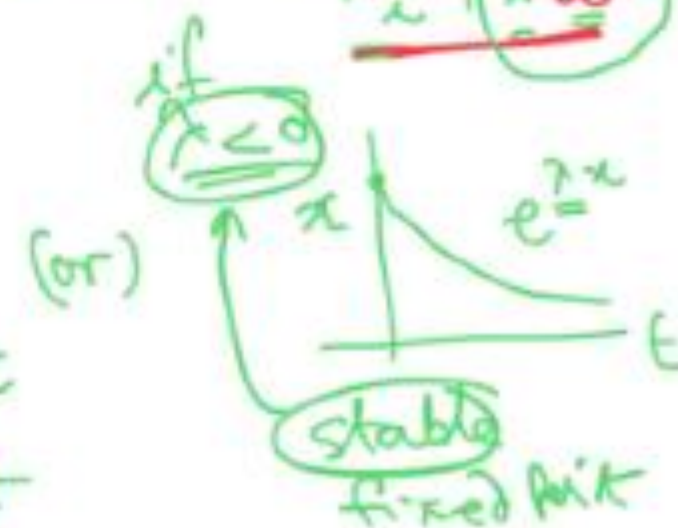
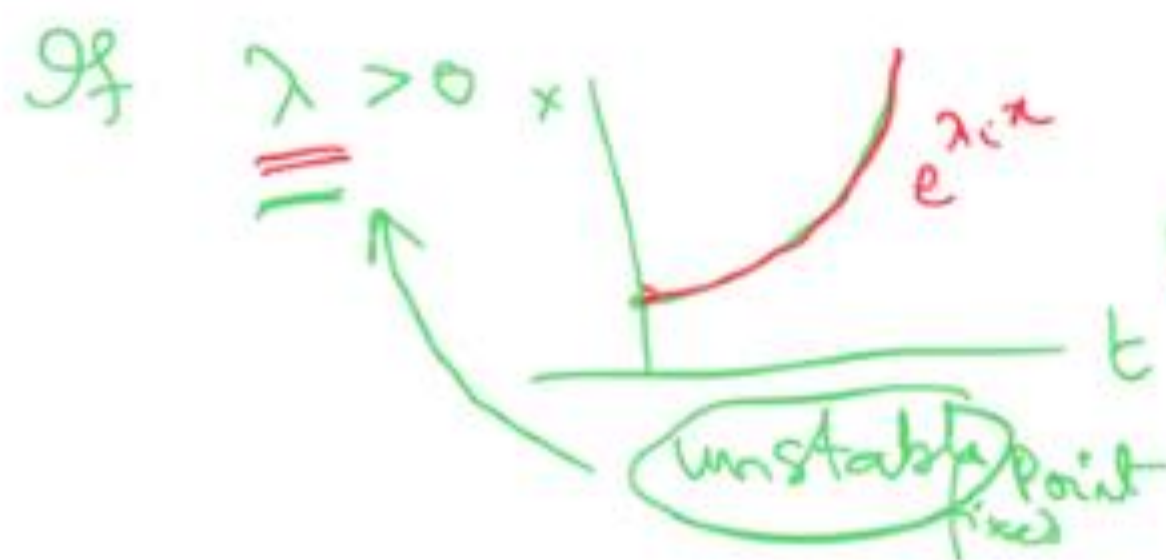
$$\frac{d\vec{x}}{dt} = \begin{pmatrix} F_u & F_w \\ G_u & G_w \end{pmatrix} \vec{x}$$

(Jacobian matrix)

Linear matrix equation.

Eigen decomposition of Jacobian.

$\rightarrow$ Eigen values. (Real + Complex)



Solution of Linear DE is

$$\vec{x}(t) = \vec{e} \exp(\lambda t)$$

Neuronal Dynamics – 4.3 Detour. Stability of fixed points

$$\tau \frac{du}{dt} = F(u, w) + RI_0$$

$$\tau_w \frac{dw}{dt} = G(u, w)$$

zoom in:

$$x = u - u_0$$

$$y = w - w_0$$

$$\tau \frac{dx}{dt} = F_u x + F_w y$$

$$\tau_w \frac{dy}{dt} = G_u x + G_w y$$

Fixed point at (u_0, w_0)

At fixed point

$$0 = F(u_0, w_0) + RI_0$$

$$0 = G(u_0, w_0)$$

$$\frac{d}{dt} \mathbf{x} = \begin{pmatrix} F_u & F_w \\ G_u & G_w \end{pmatrix} \mathbf{x},$$

Neuronal Dynamics – 4.3 Detour. Stability of fixed points

Linear matrix equation

$$\frac{d}{dt}x = \begin{pmatrix} F_u & F_w \\ G_u & G_w \end{pmatrix} x ,$$

Search for solution

$$x(t) = e \exp(\lambda t)$$

Two solution with Eigenvalues λ_+, λ_-

$$\lambda_+ + \lambda_- = F_u + G_w$$

$$\lambda_+ \lambda_- = F_u G_w - F_w G_u$$

Neuronal Dynamics – 4.3 Detour. Stability of fixed points

Linear matrix equation

$$\frac{d}{dt}x = \begin{pmatrix} F_u & F_w \\ G_u & G_w \end{pmatrix} x$$

Search for solution

$$x(t) = e \exp(\lambda t)$$

Two solution with Eigenvalues λ_+, λ_-

$$\lambda_+ + \lambda_- = F_u + G_w$$

$$\lambda_+ \lambda_- = F_u G_w - F_w G_u$$

Stability requires:

$$\lambda_+ < 0 \quad \text{and} \quad \lambda_- < 0$$



$$F_u + G_w < 0$$

and

$$F_u G_w - F_w G_u > 0$$



Neuronal Dynamics – 4.3 Detour. Stability of fixed points

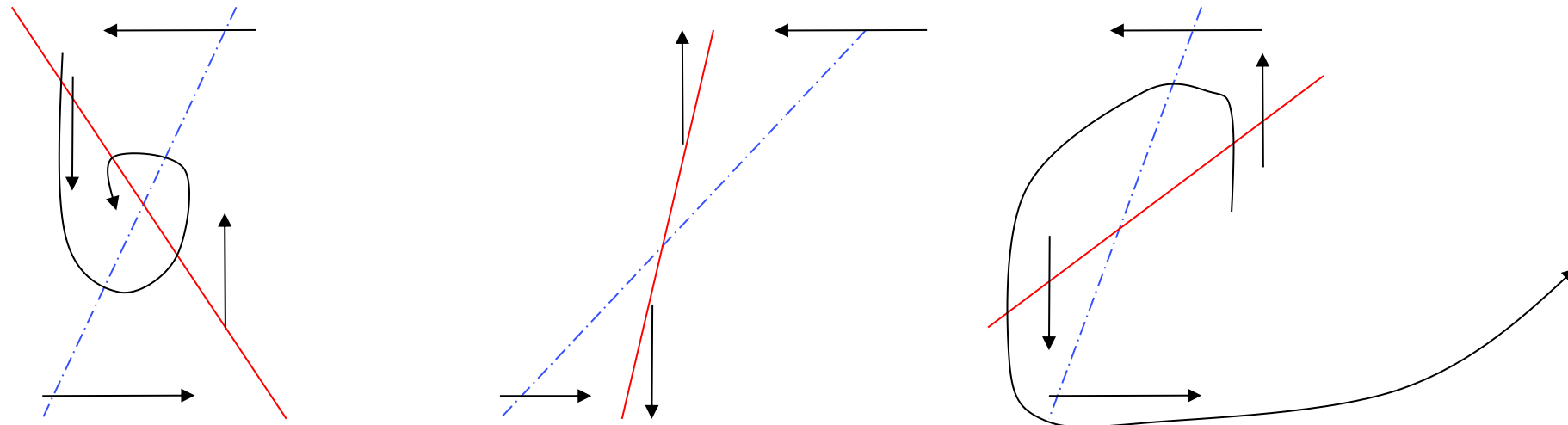
stimulus



$$\tau \frac{du}{dt} = au - w + I_0$$

$$\tau_w \frac{dw}{dt} = cu - w$$

$\lambda_{+/-} =$



w

$$\frac{dw}{dt} = 0$$

$$I(t) = I_0$$

u

$$\frac{du}{dt} = 0$$

Neuronal Dynamics – 4.3 Detour. Stability of fixed points

2-dimensional equation

$$\tau \frac{du}{dt} = F(u, w) + RI_0$$

$$\tau_w \frac{dw}{dt} = G(u, w)$$

Stability characterized
by Eigenvalues of
linearized equations

$$\frac{d}{dt} \mathbf{x} = \begin{pmatrix} F_u & F_w \\ G_u & G_w \end{pmatrix} \mathbf{x}$$

NOW Exercise 2.1: Stability of Fixed Point in 2D

$$\frac{du}{dt} = \alpha u - w$$

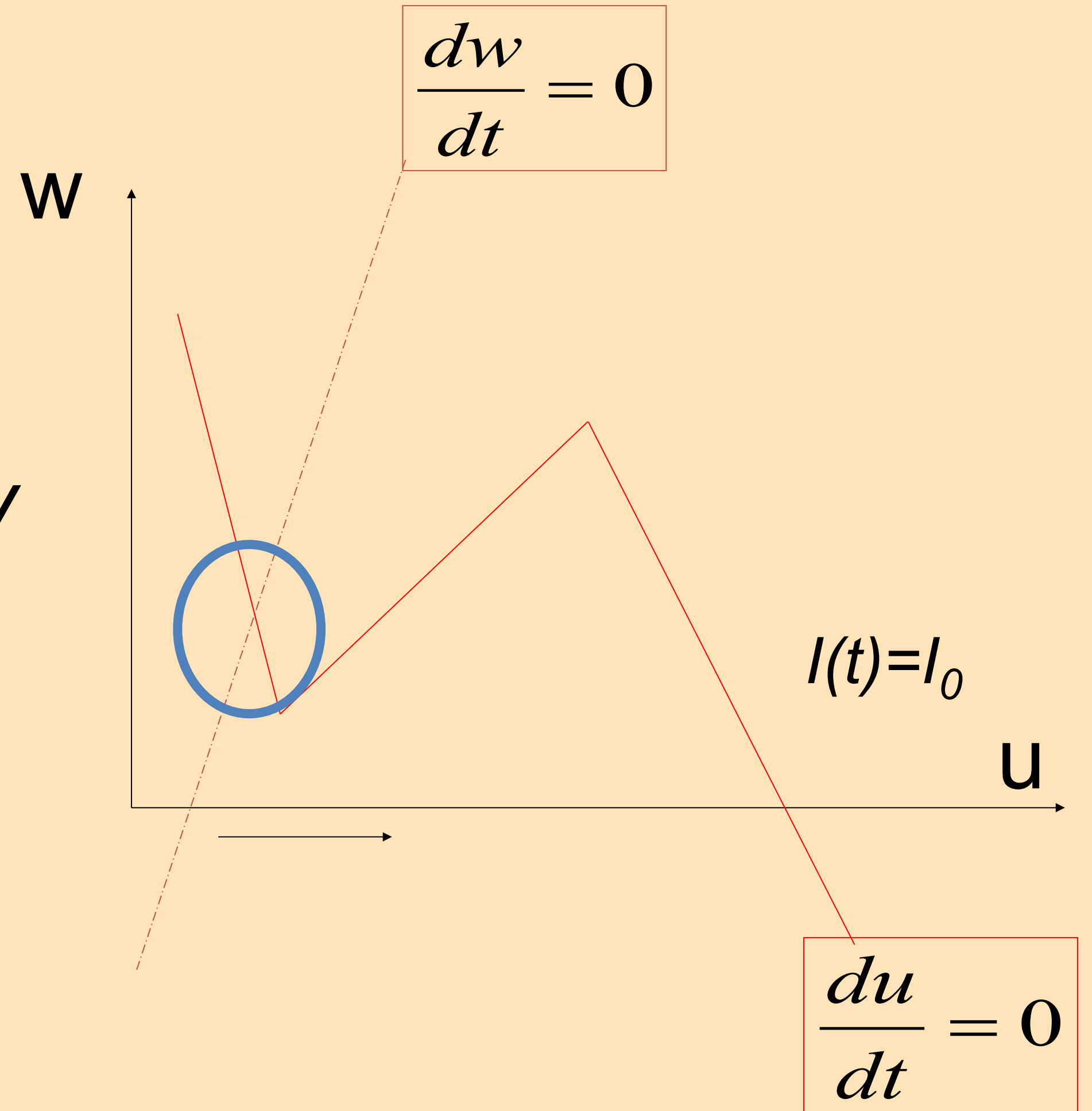
$$\frac{dw}{dt} = \beta u - w$$

- calculate *stability*

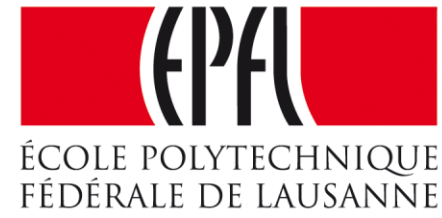
- compare

$$\frac{dx}{dt} = -\frac{x}{\tau}$$

Exercises:
2.1, 2.2



Week 4: Reducing Detail – 2D models-Adding Detail



Biological Modeling of Neural Networks

Week 4

– Reducing detail

- **Adding detail**

Wulfram Gerstner

EPFL, Lausanne, Switzerland



✓ 4.1 From Hodgkin-Huxley to 2D

✓ 4.2 Phase Plane Analysis

4.3 Analysis of a 2D Neuron Model

4.4 Type I and II Neuron Models
- limit cycles

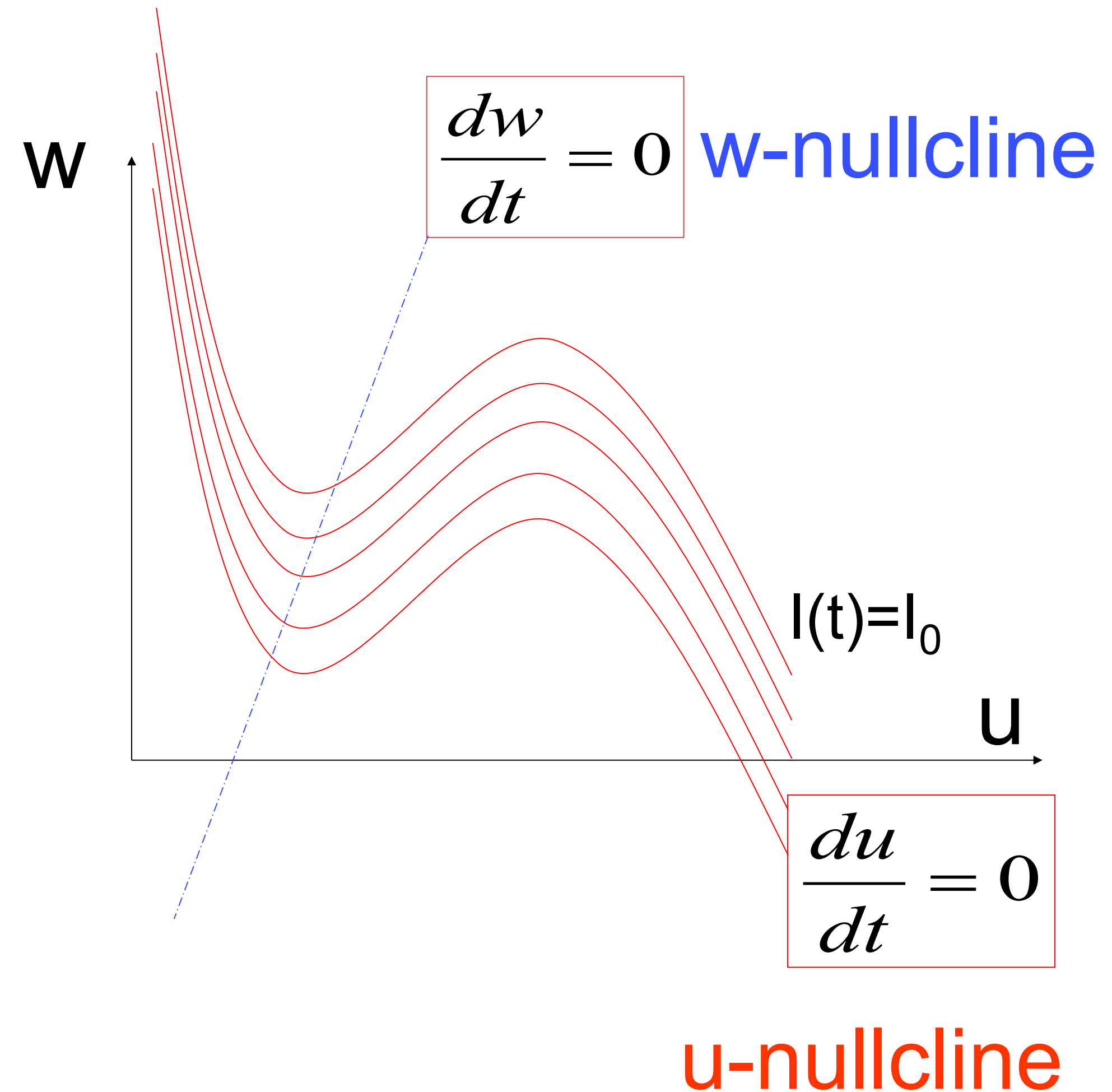
4.4. 2 dimensional Neuron Models

$$\tau \frac{du}{dt} = F(u, w) + I(t)$$

stimulus
↓

$$\tau_w \frac{dw}{dt} = G(u, w)$$

apply constant stimulus I_0



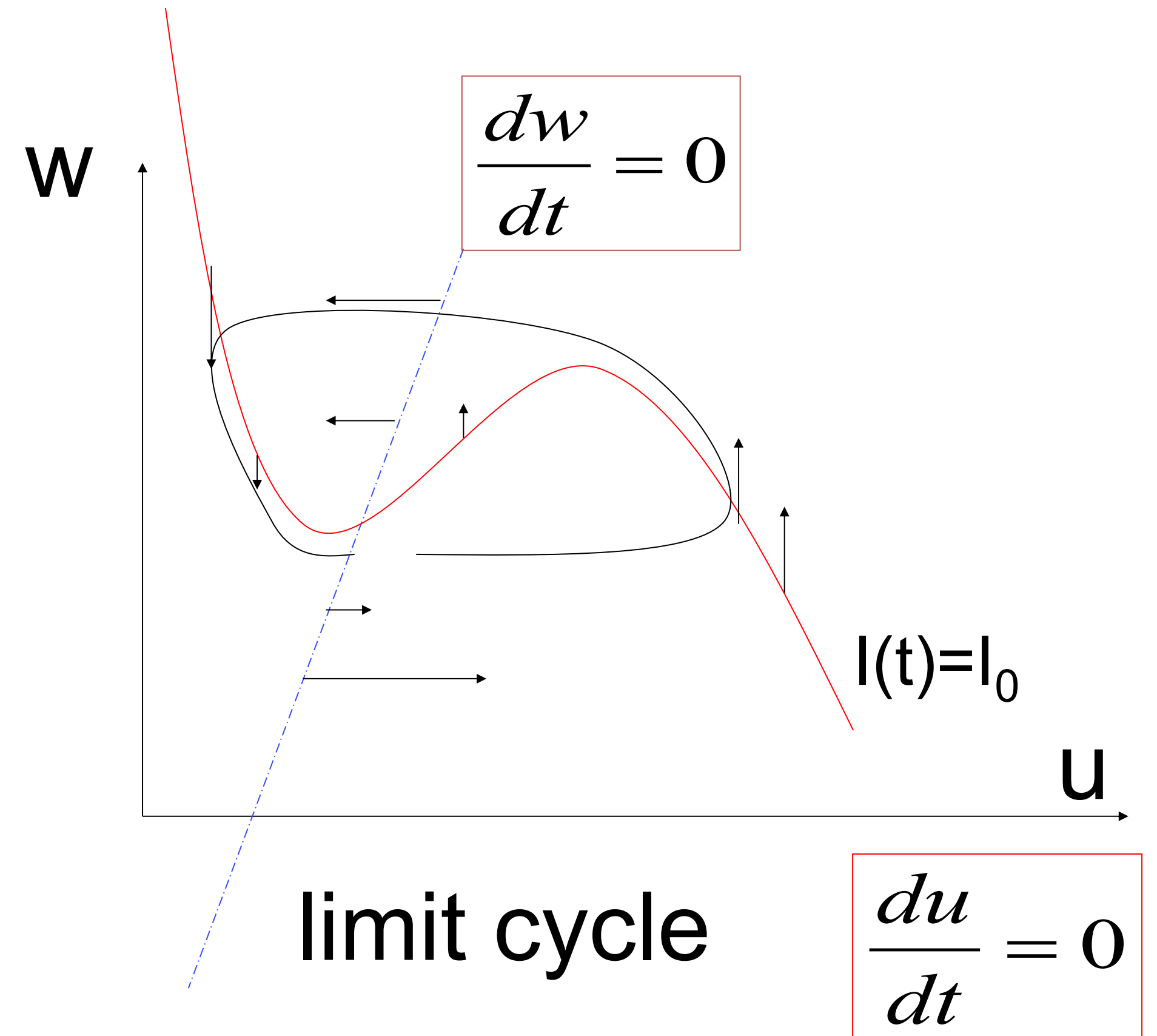
4.4. FitzHugh Nagumo Model – limit cycle

$$\tau \frac{du}{dt} = F(u, w) + I(t)$$

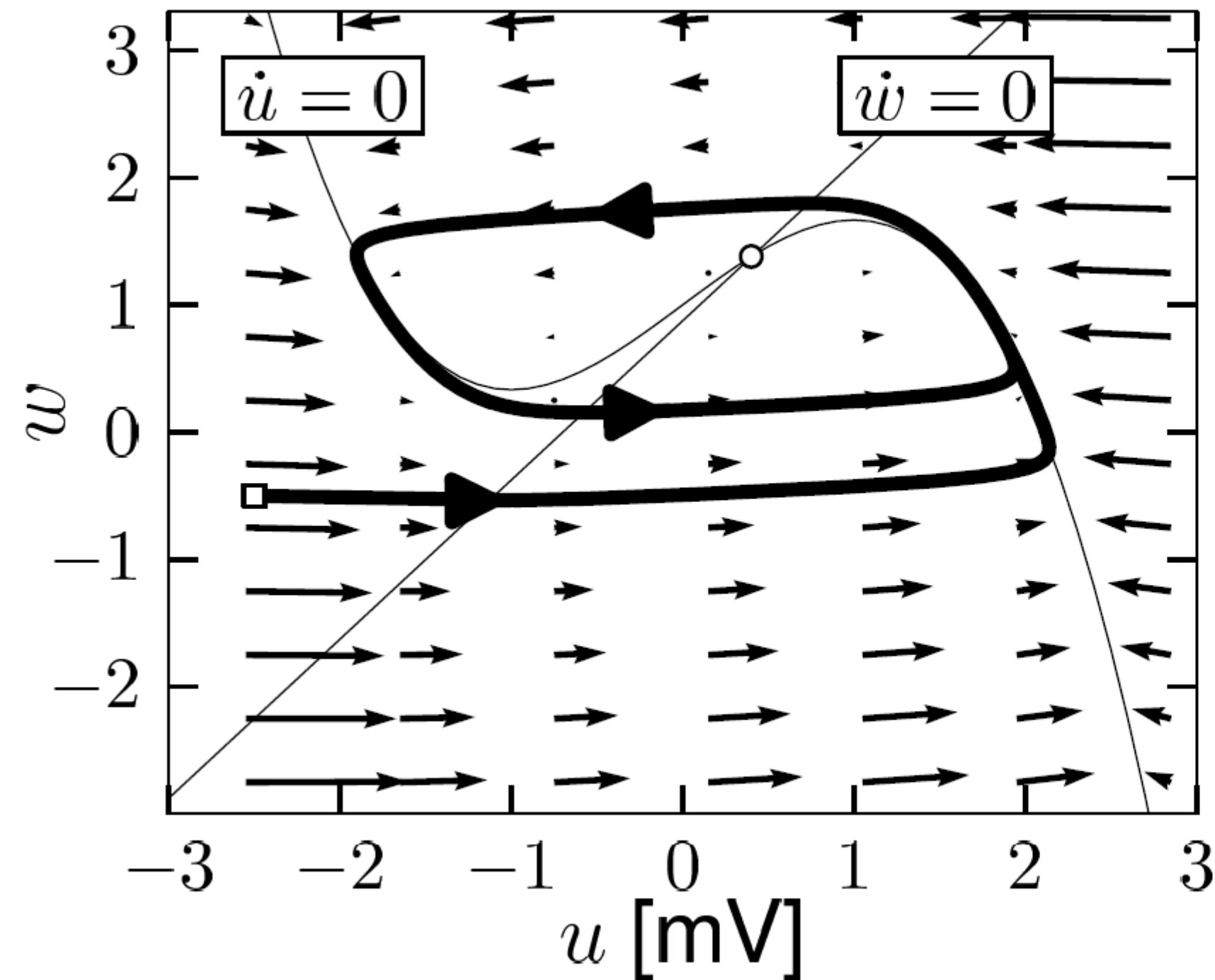
stimulus
↓

$$\tau_w \frac{dw}{dt} = G(u, w)$$

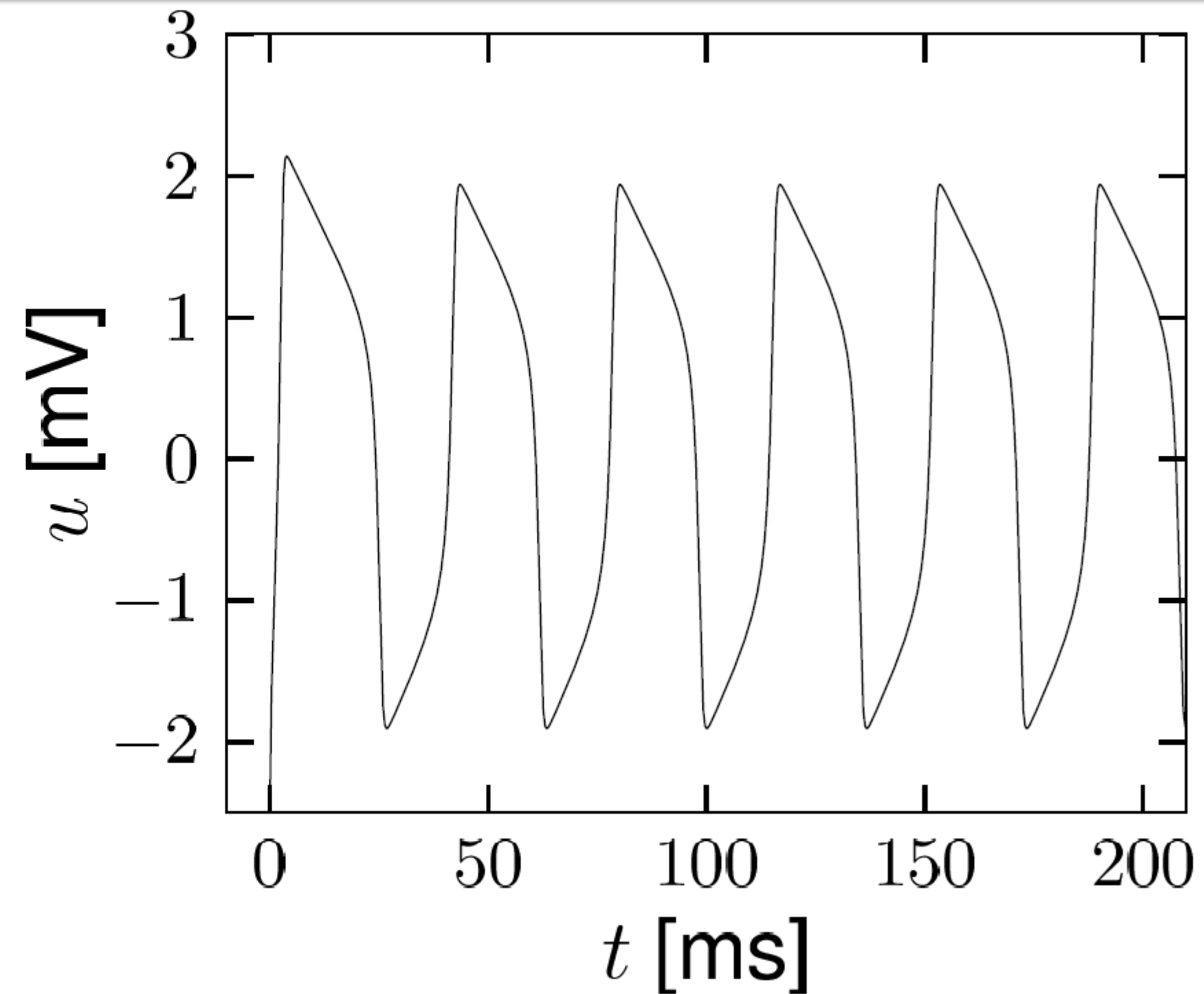
- unstable fixed point
 - closed boundary
with arrows pointing inside
- limit cycle



Neuronal Dynamics – 4.4. Limit Cycle



D



- unstable fixed point in 2D
- bounding box with inward flow
→ limit cycle (*Poincare Bendixson*)

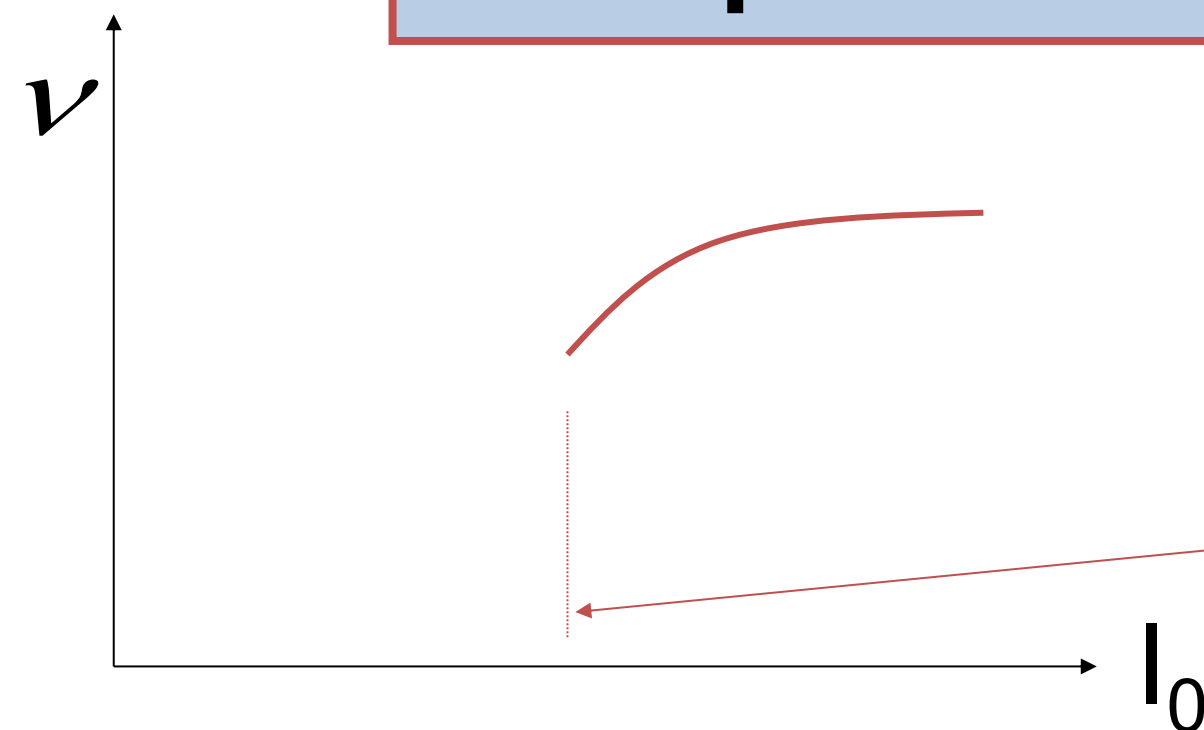
*Image: Neuronal Dynamics,
Gerstner et al.,
Cambridge Univ. Press (2014)*

Type II Model constant input

$$\tau \frac{du}{dt} = F(u, w) + I(t)$$

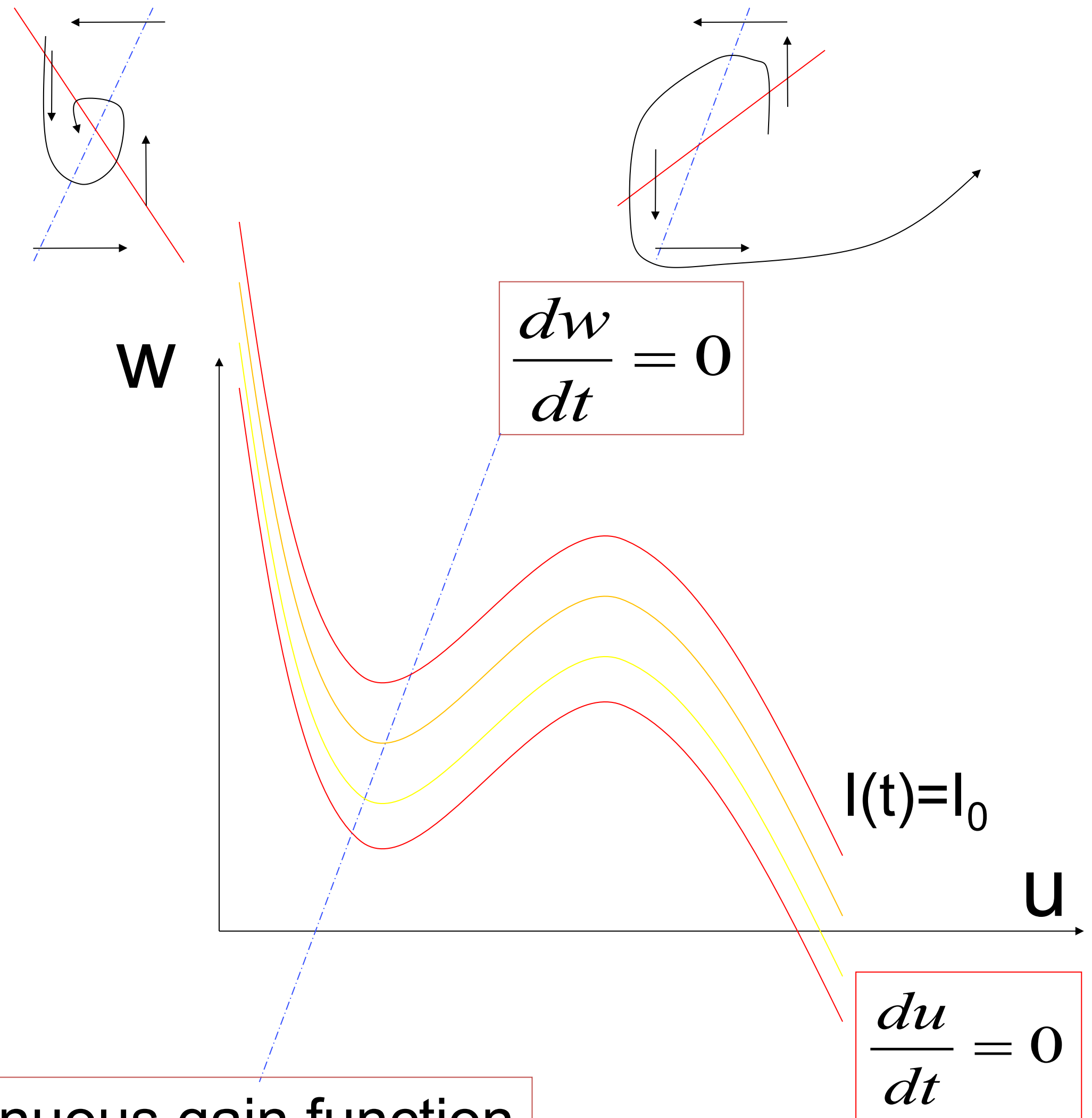
$$\tau_w \frac{dw}{dt} = G(u, w)$$

Hopf bifurcation



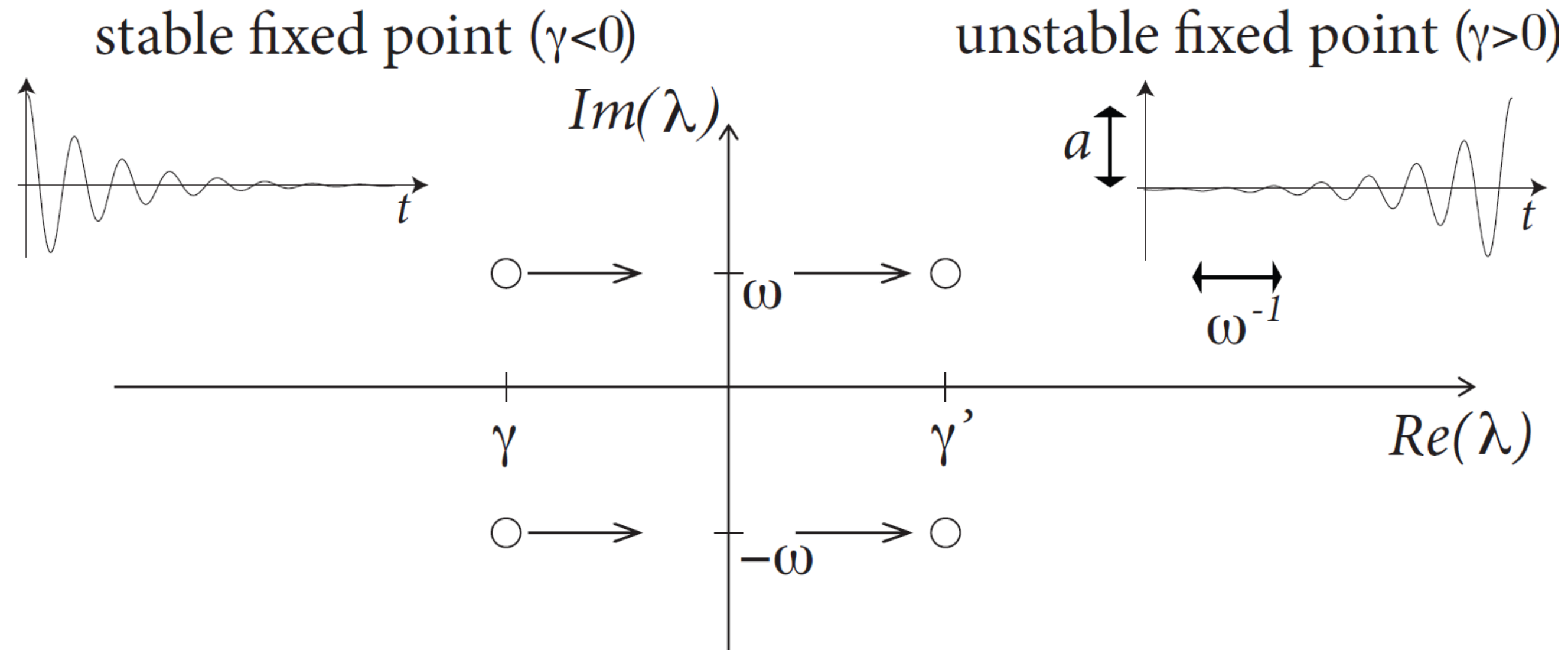
Discontinuous gain function

Stability lost $\rightarrow$ oscillation with finite frequency

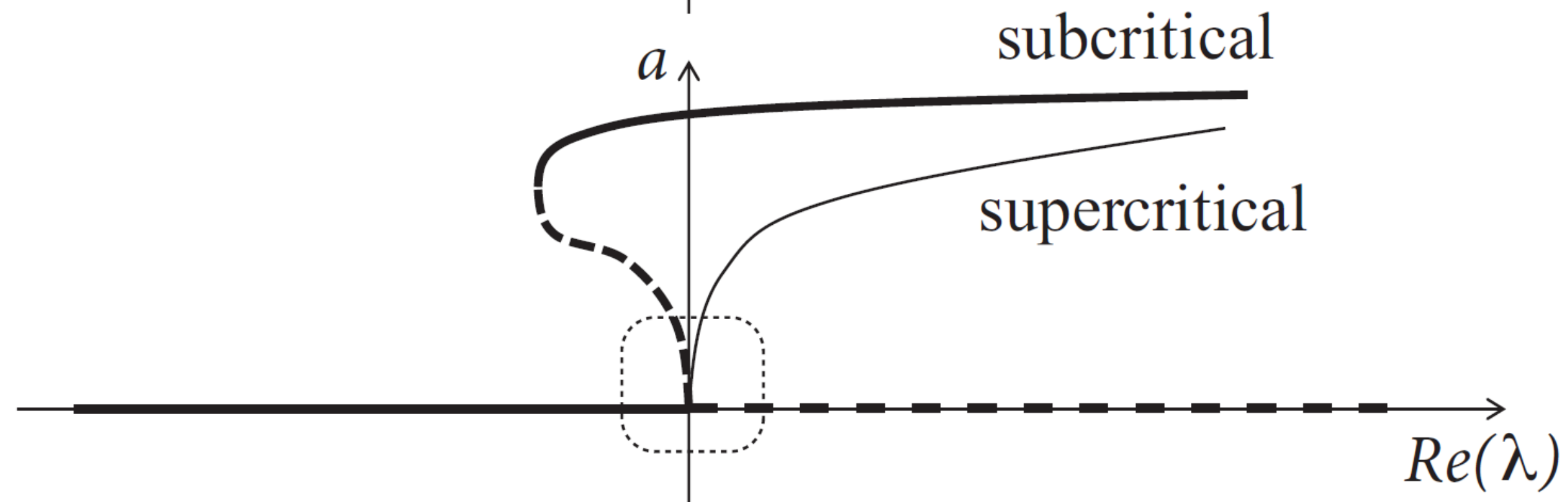
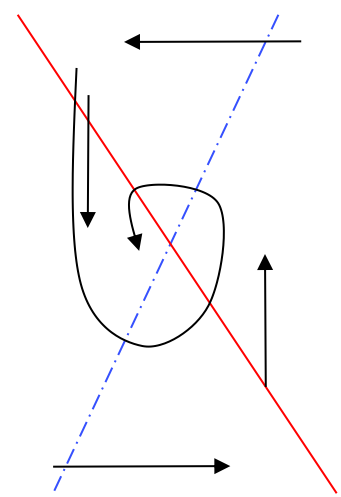


Neuronal Dynamics – 4.4. Hopf bifurcation

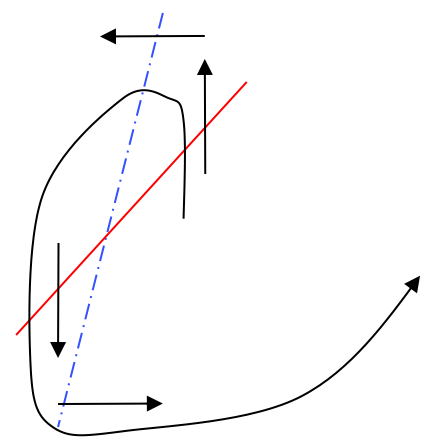
$$\lambda = \gamma + i\omega$$



$$\gamma < 0$$

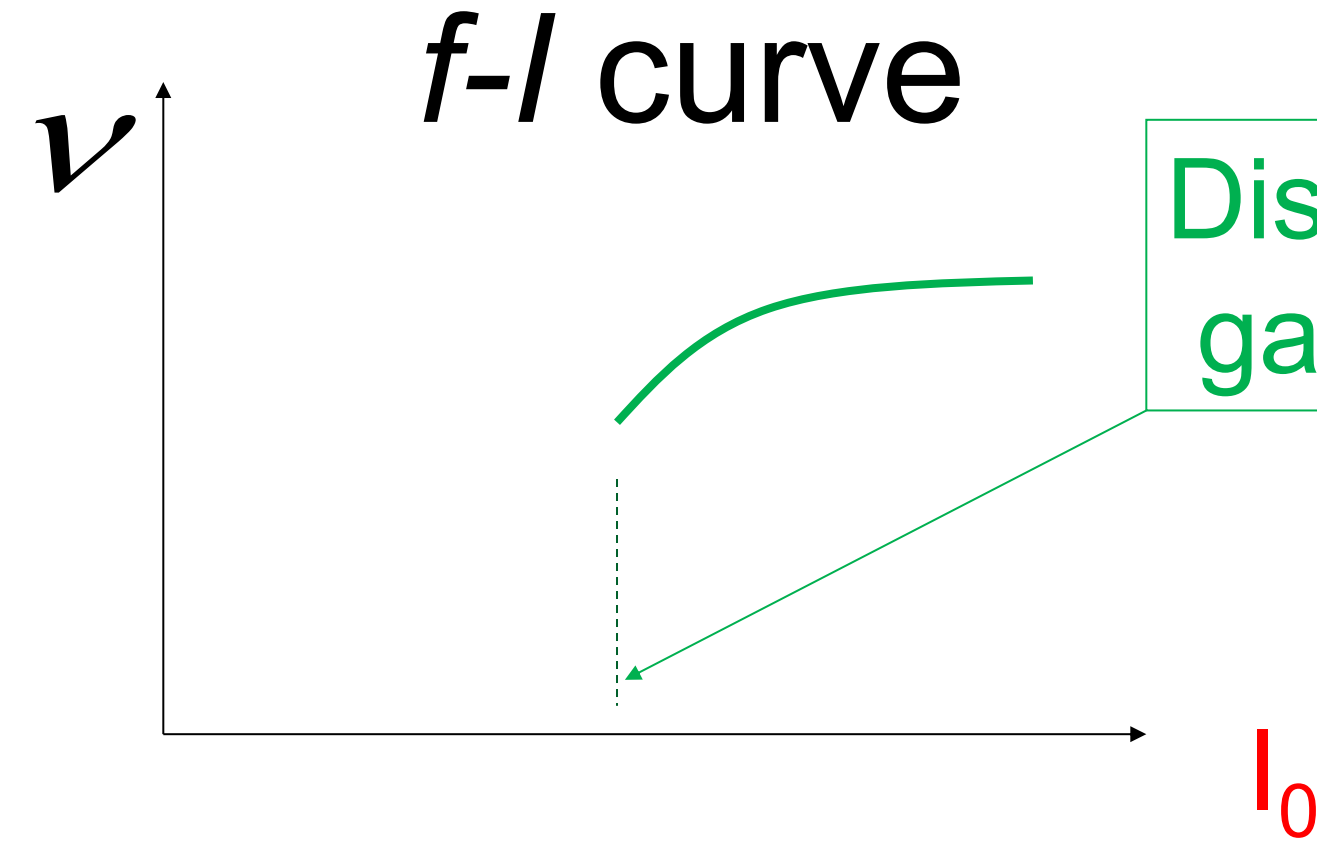
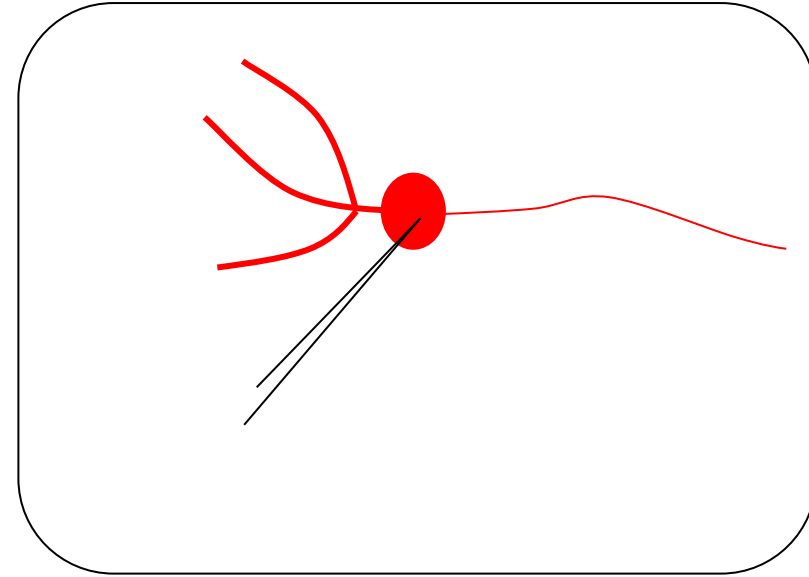
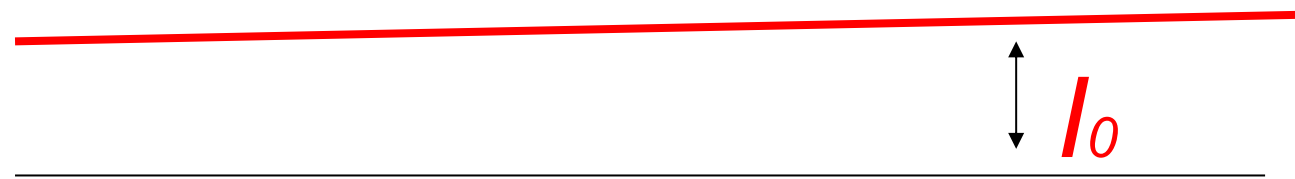


$$\gamma > 0$$

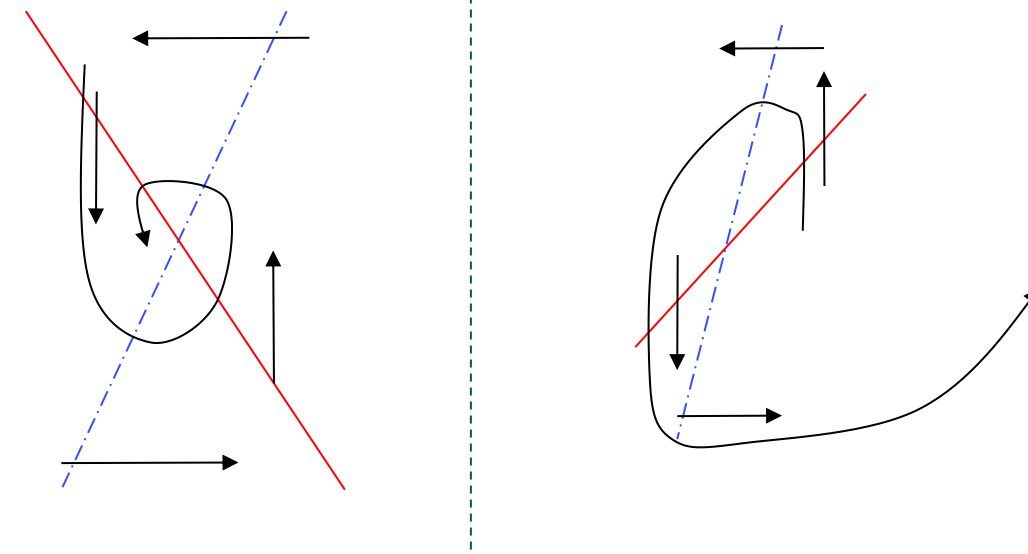


Neuronal Dynamics – 4.4. Hopf bifurcation: f - I -curve

ramp input/
constant input



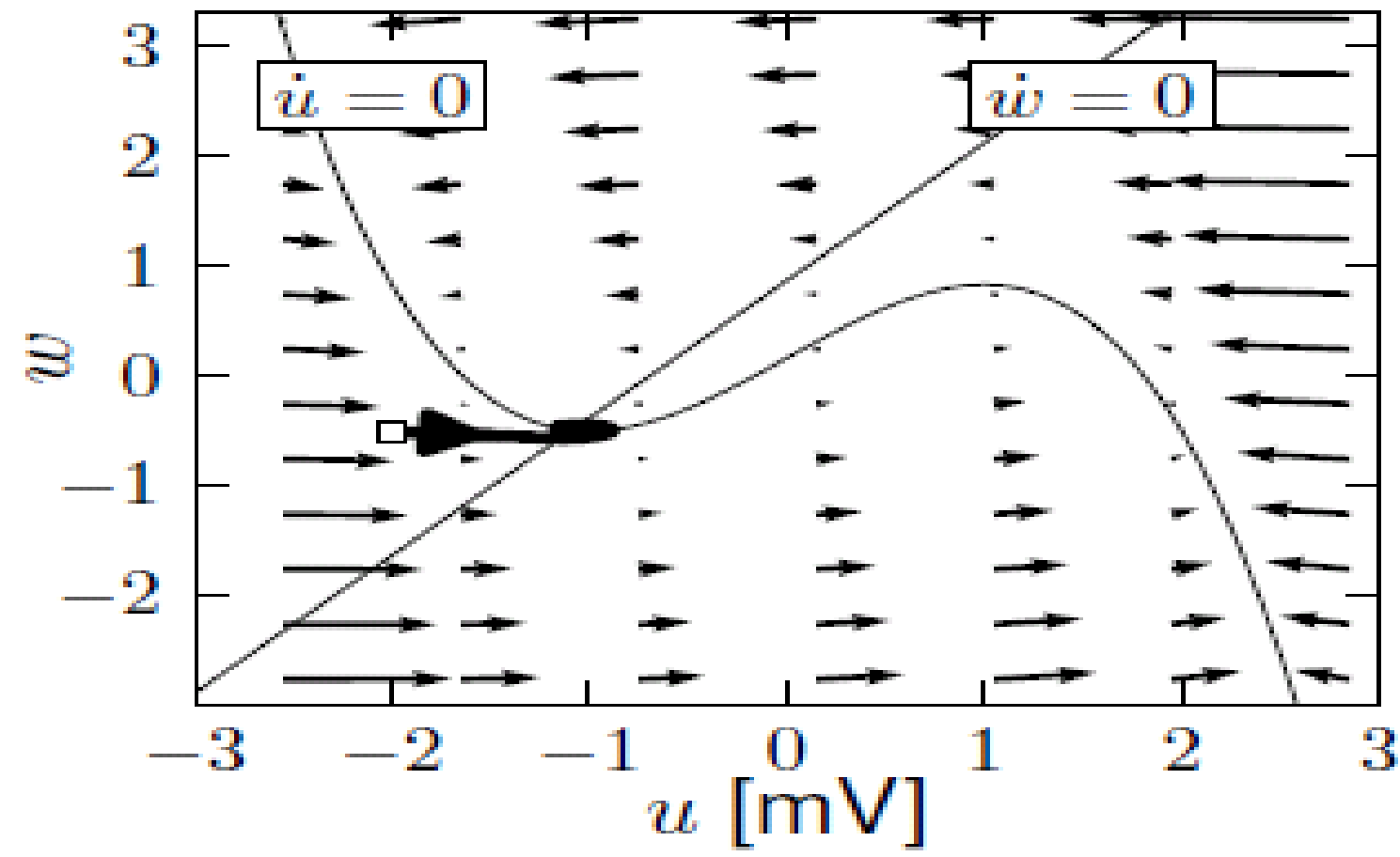
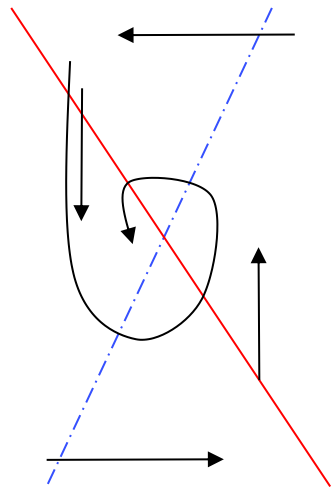
Discontinuous
gain function: Type II



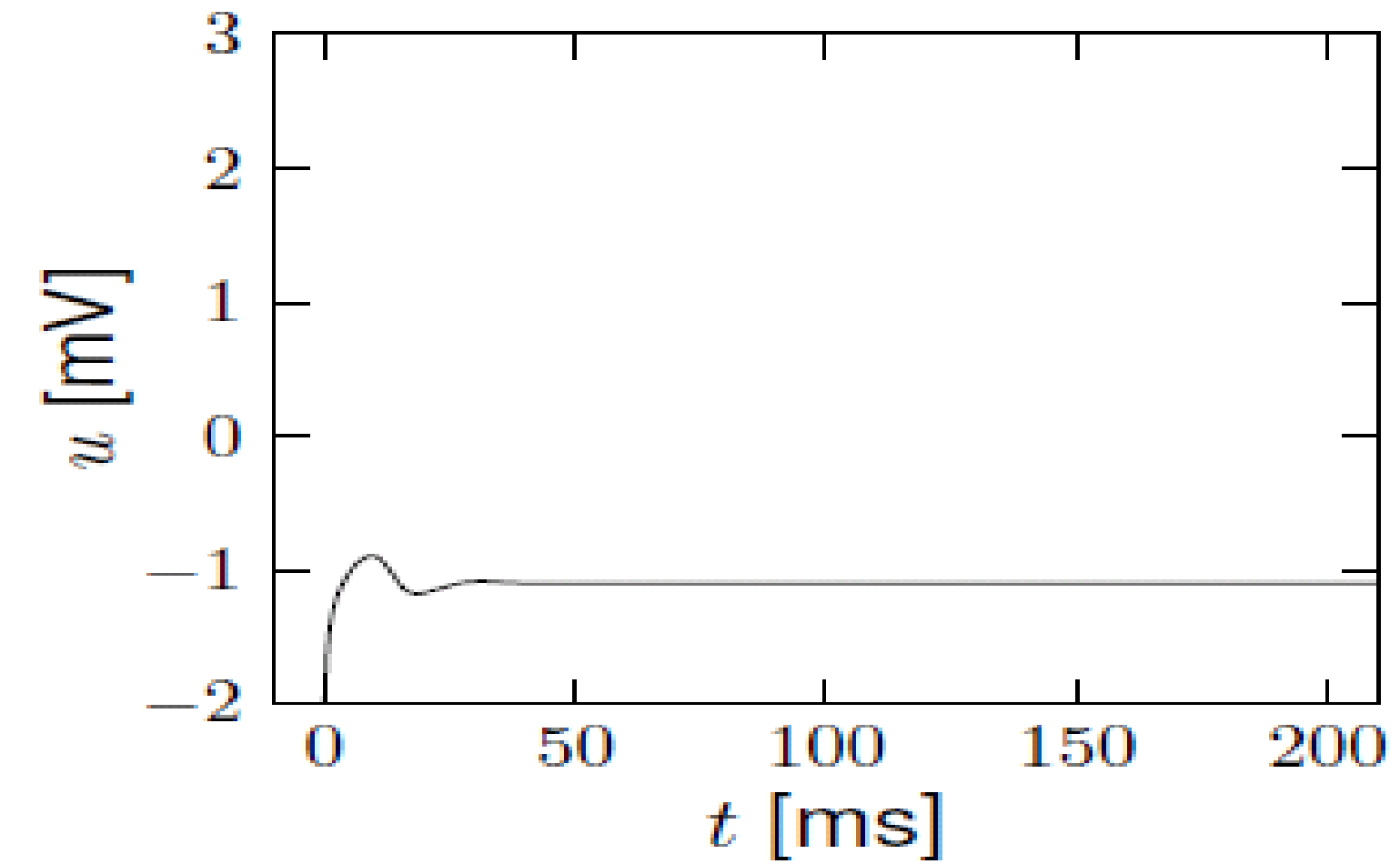
Stability lost $\rightarrow$ oscillation with finite frequency

FitzHugh-Nagumo: Type II Model – Hopf bifurcation

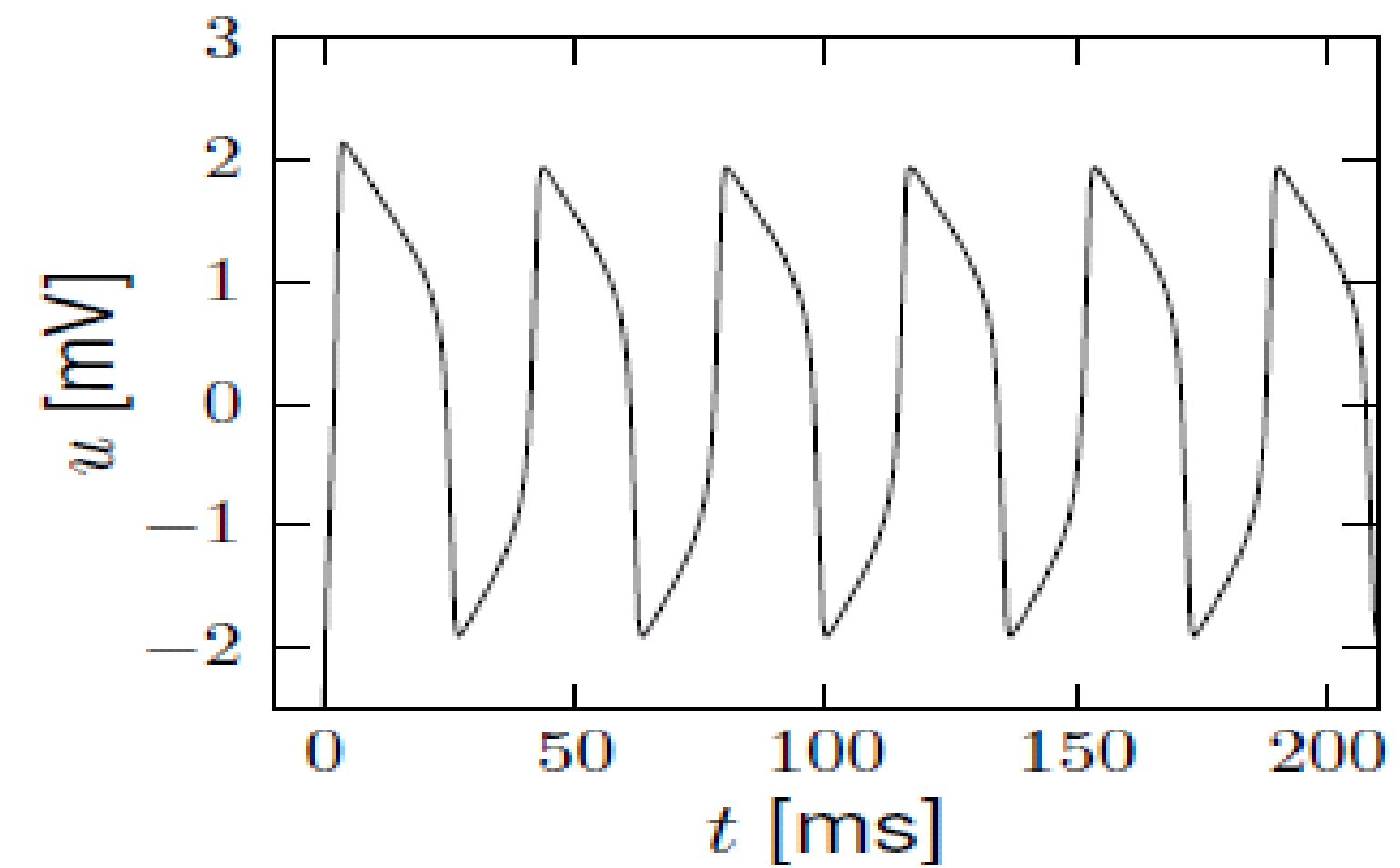
$I=0$



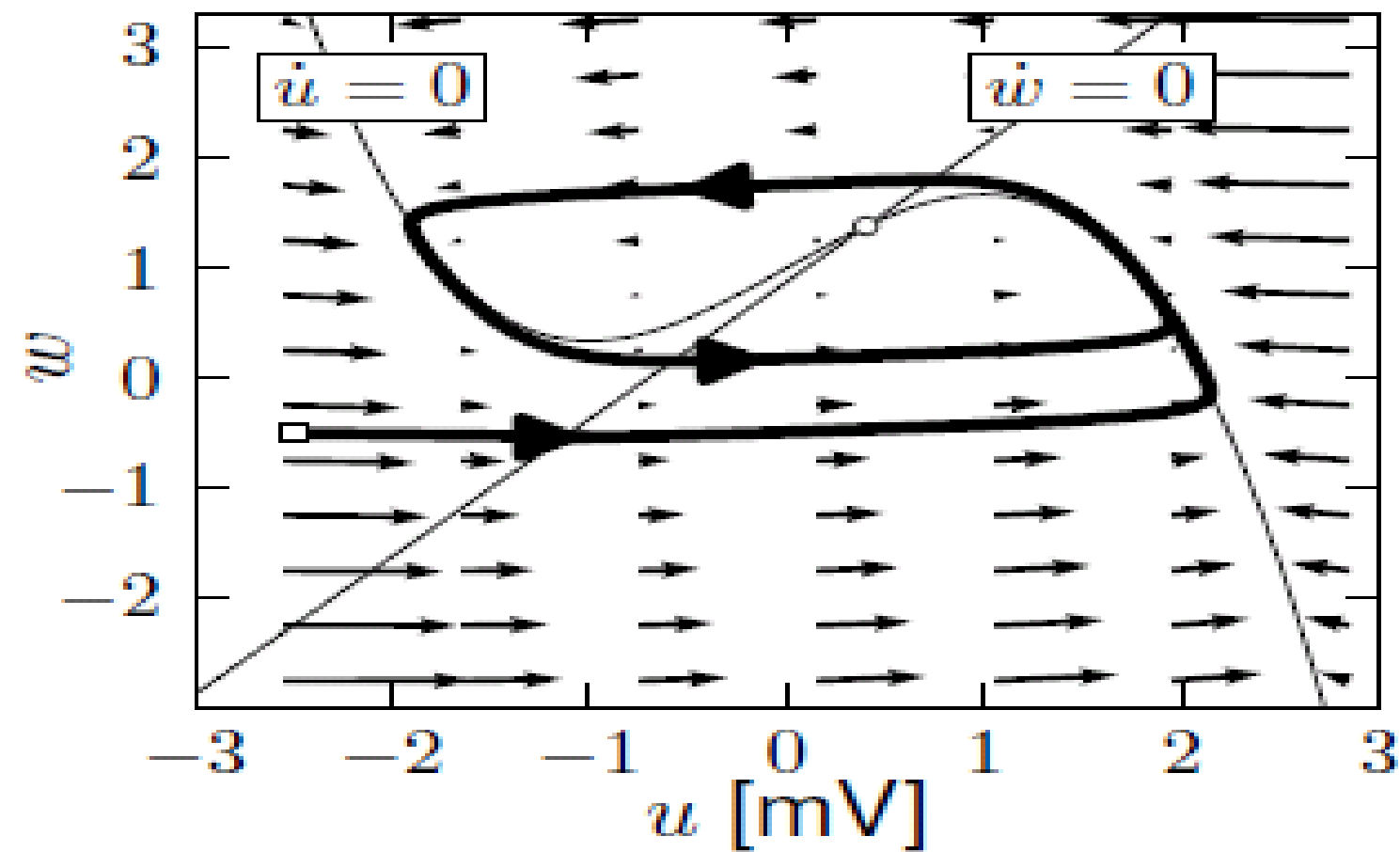
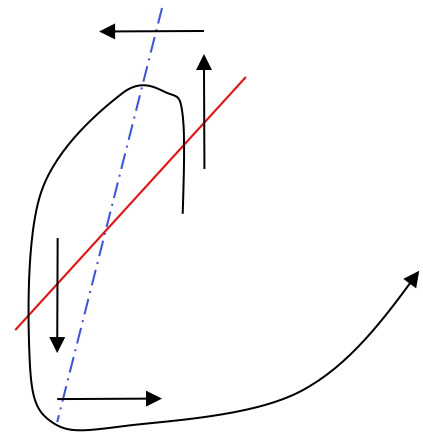
B



D

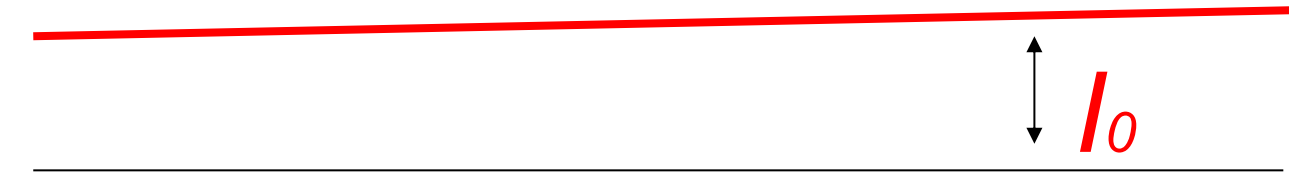


$I > I_c$



Neuronal Dynamics – 4.4. Type I and II Neuron Models

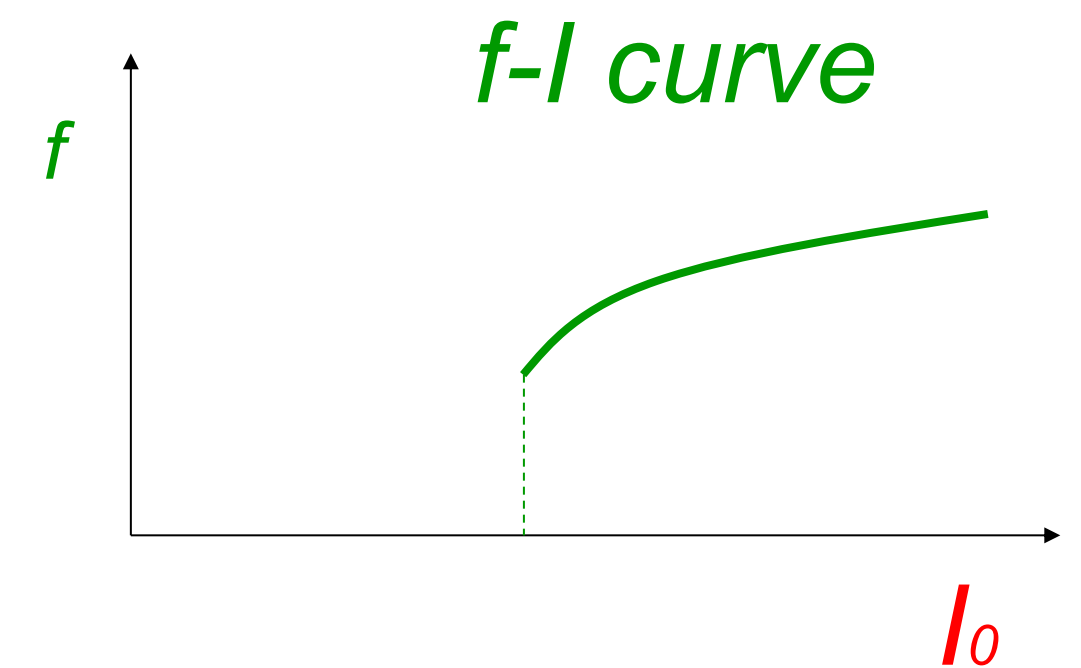
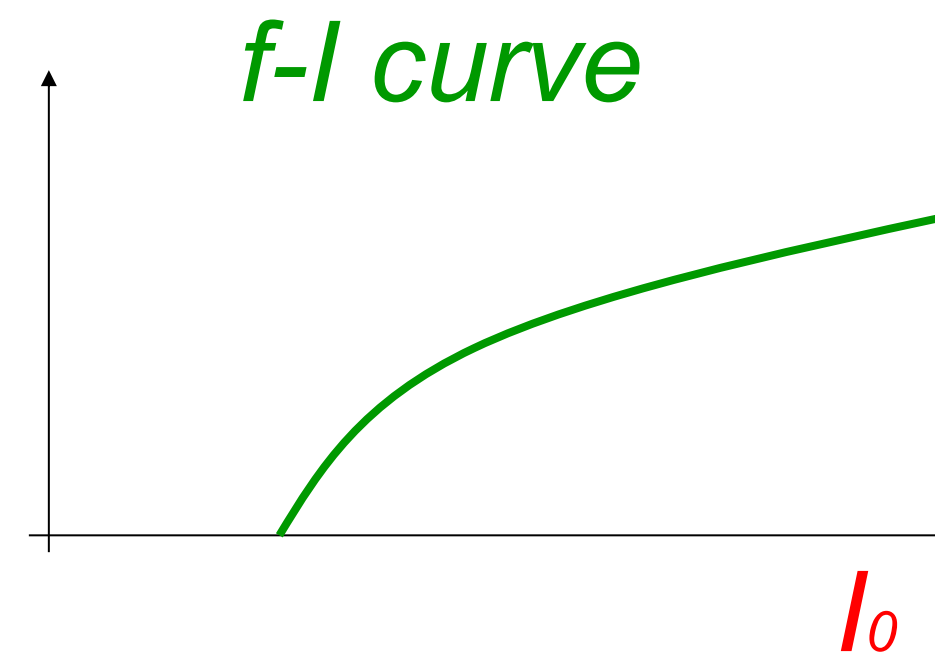
ramp input/
constant input



neuron

Now:
Type I model

Type I and type II models



Neuronal Dynamics – 4.4. Type I and II Neuron Models

type I Model: 3 fixed points

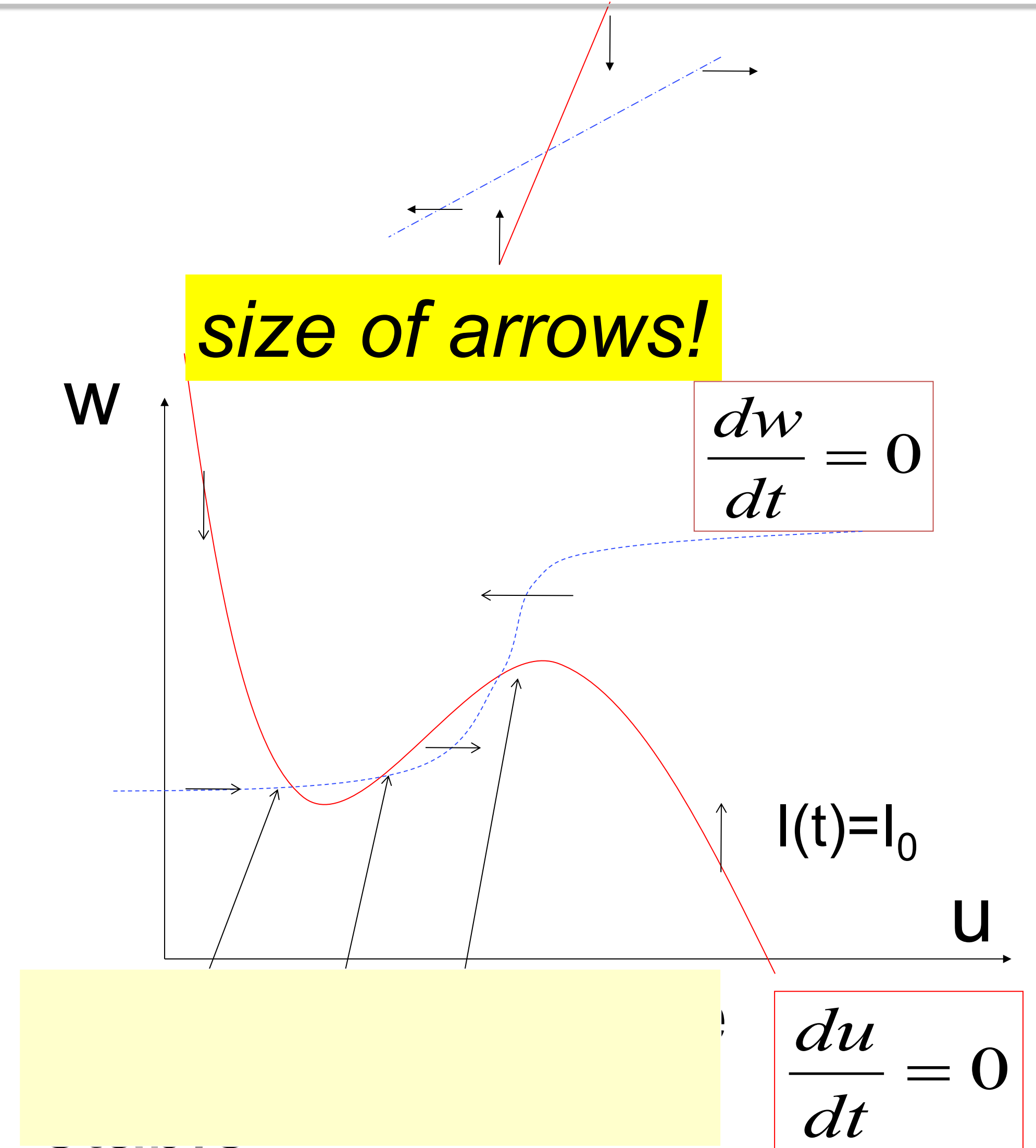
stimulus

$$\tau \frac{du}{dt} = F(u, w) + I(t)$$

$$\tau_w \frac{dw}{dt} = G(u, w)$$

apply constant stimulus I_0

Saddle-node bifurcation



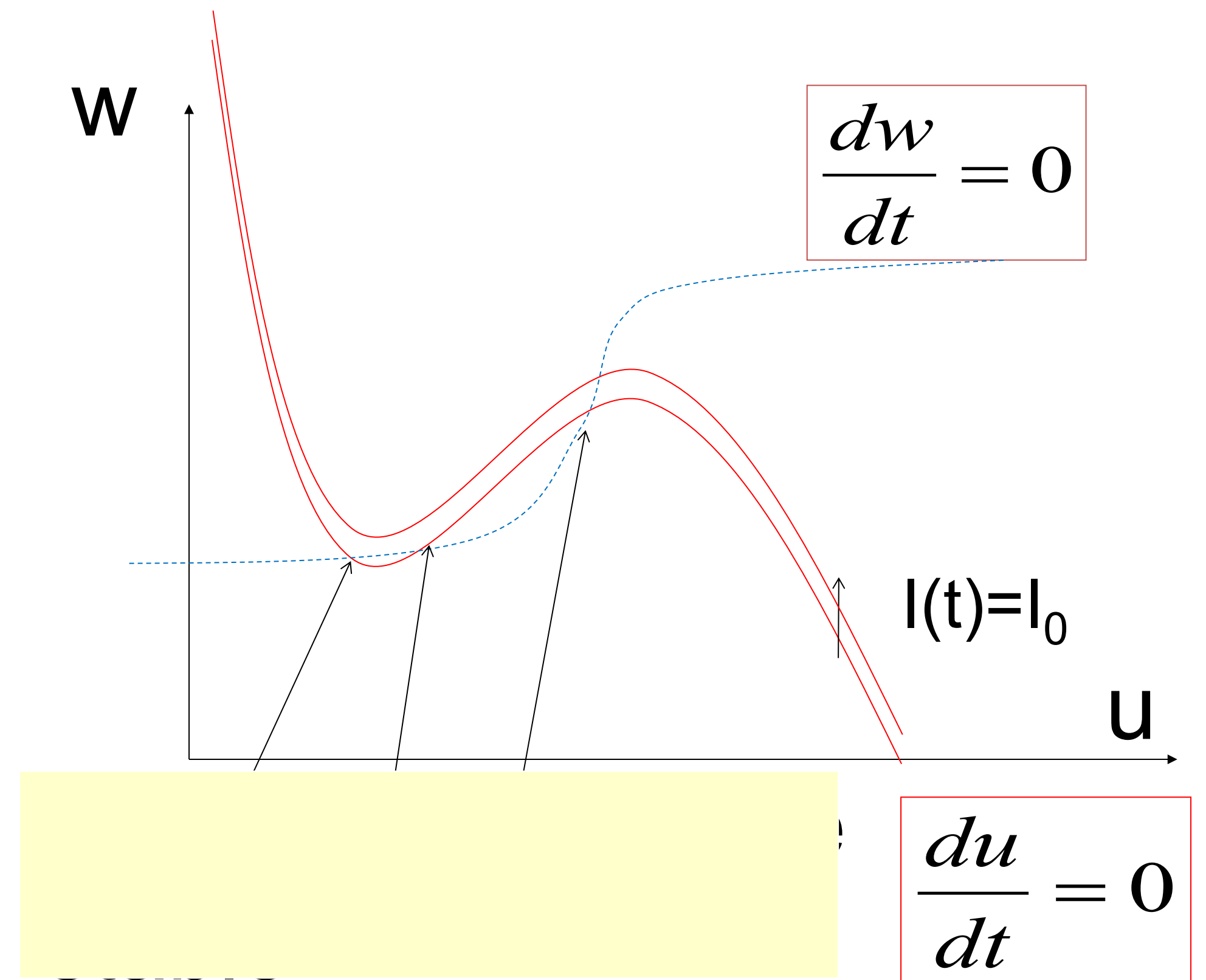
Saddle-node bifurcation

stimulus



$$\tau \frac{du}{dt} = F(u, w) + I(t)$$

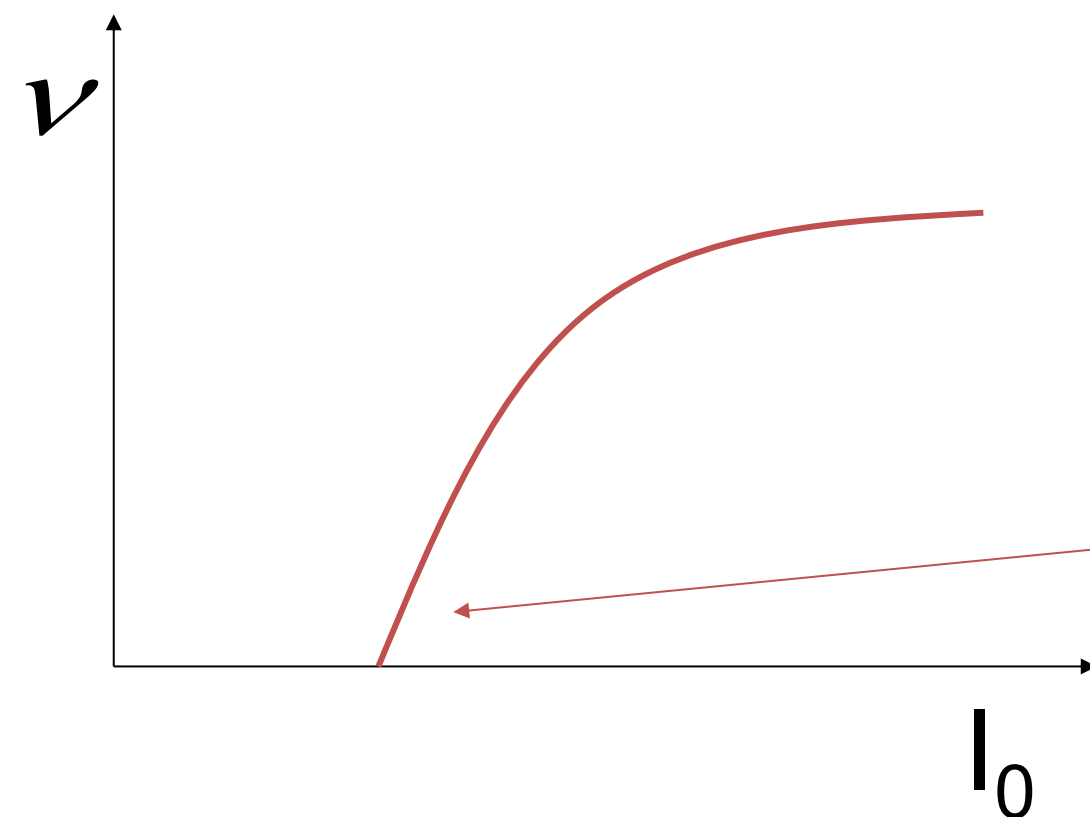
$$\tau_w \frac{dw}{dt} = G(u, w)$$



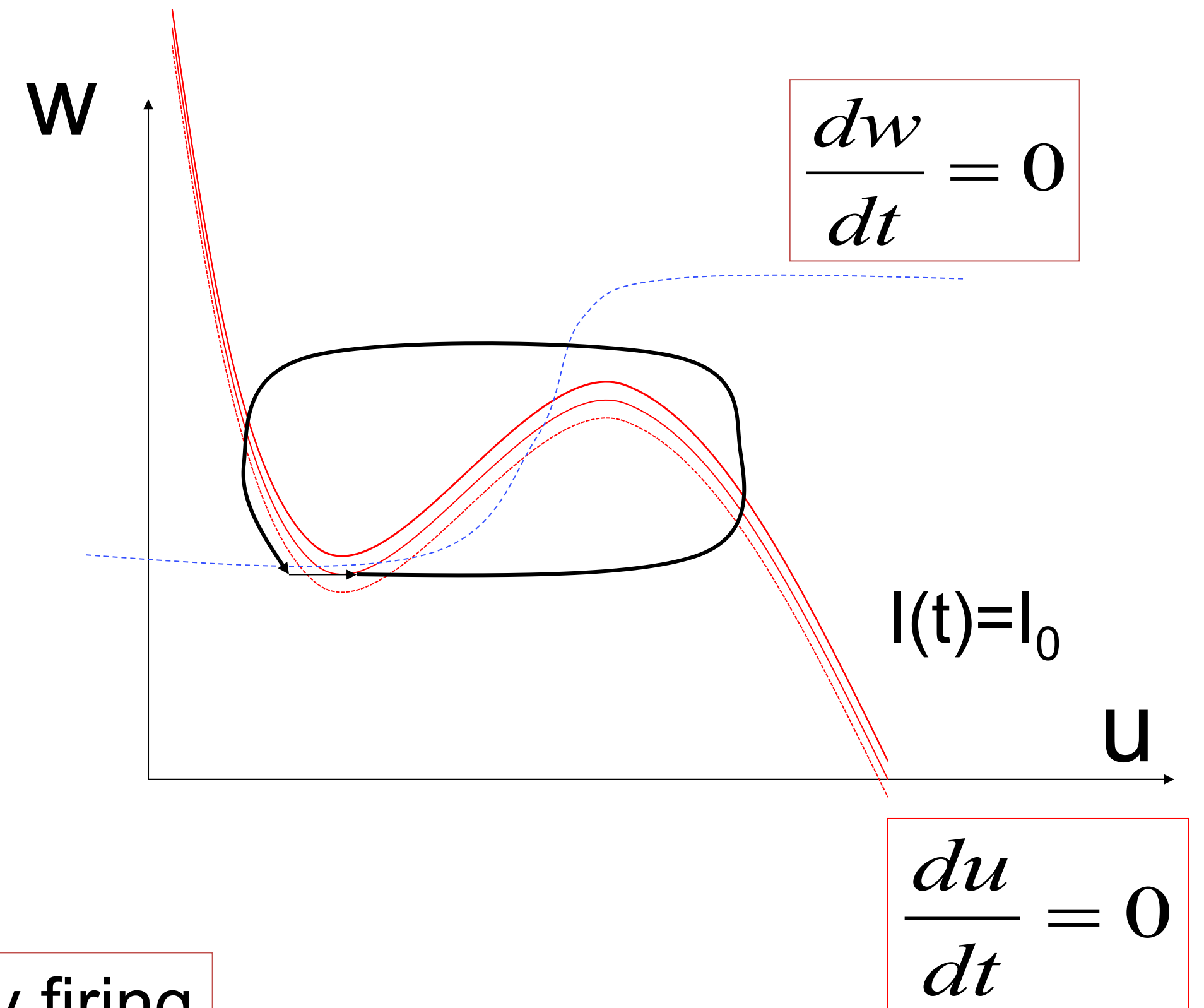
type I Model – constant input

$$\tau \frac{du}{dt} = F(u, w) + \overset{\text{stimulus}}{\downarrow} I(t)$$

$$\tau_w \frac{dw}{dt} = G(u, w)$$



Low-frequency firing



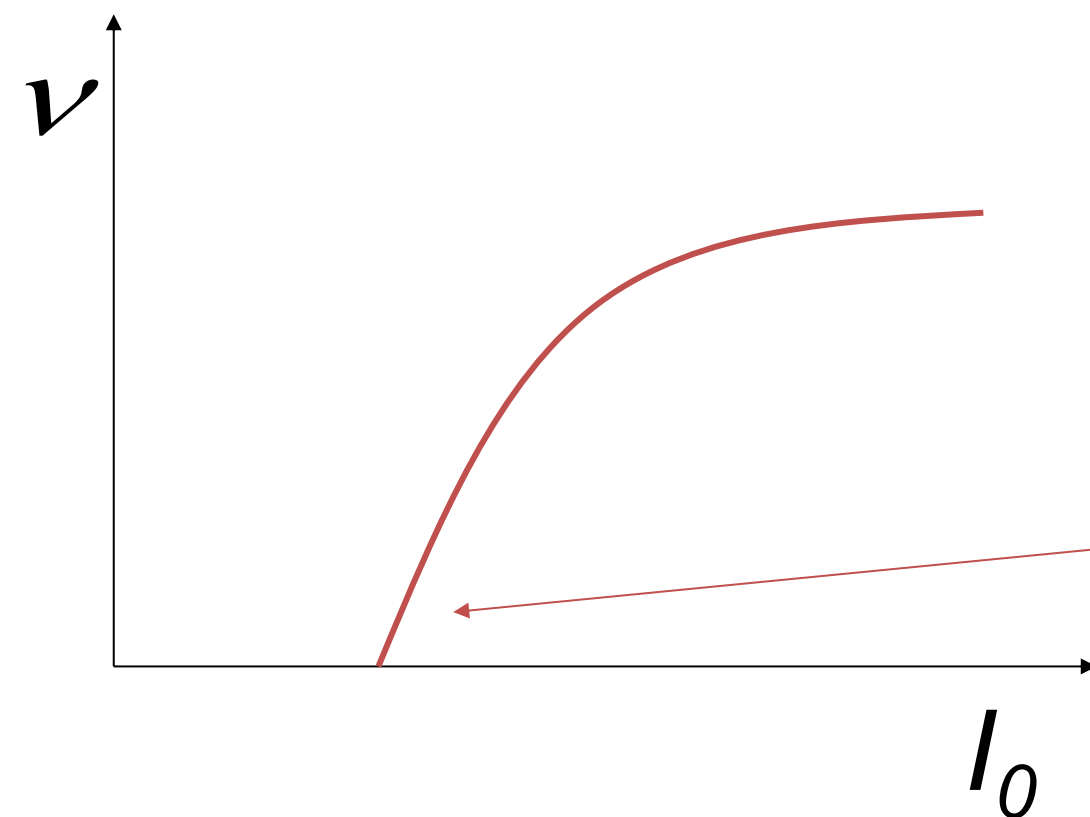
type I Model – Morris-Lecar: constant input

stimulus

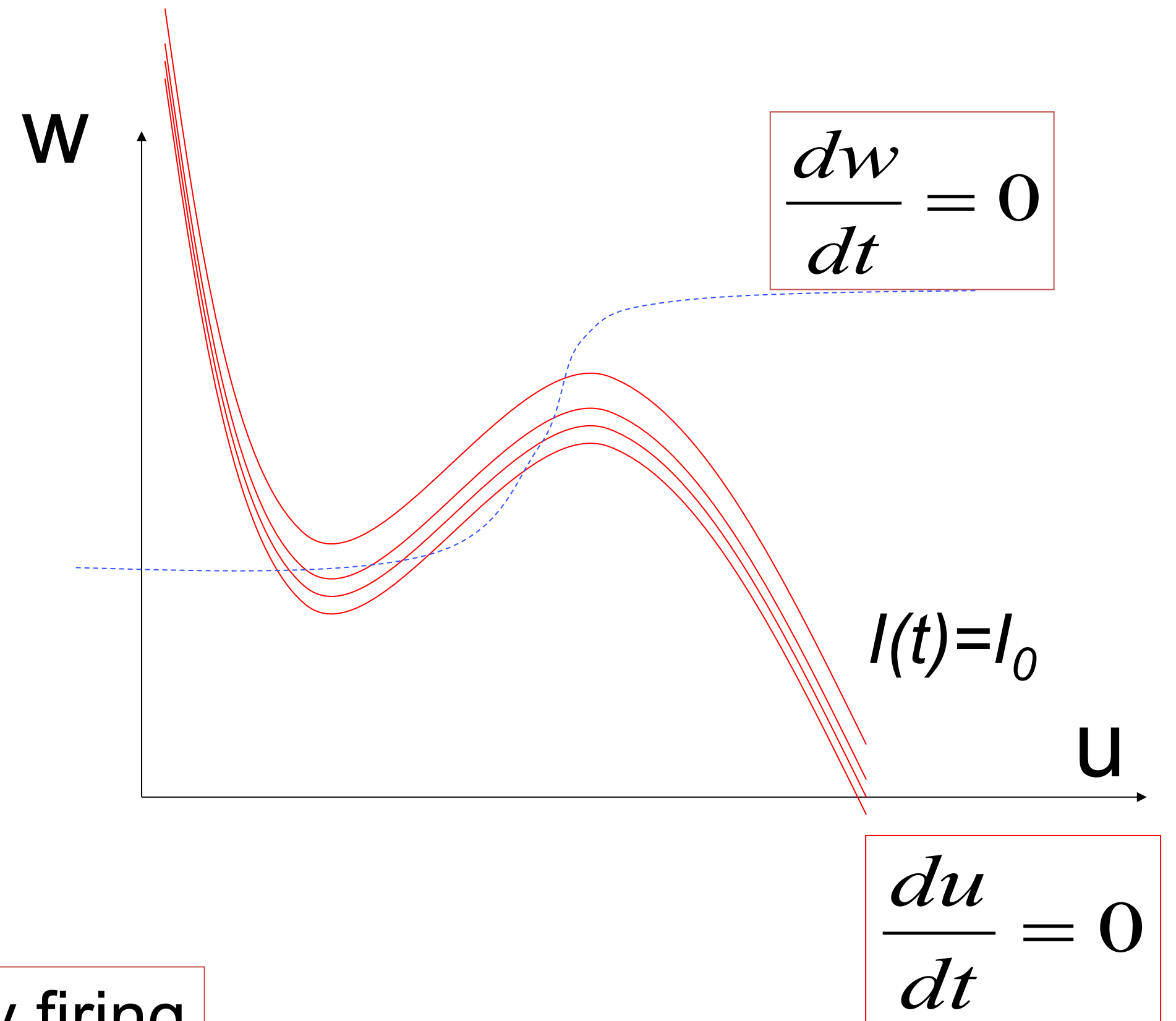
$$\tau \frac{du}{dt} = F(u, w) + I(t)$$

$$\frac{dw}{dt} = -\frac{w - w_0(u)}{\tau_{eff}(u)}$$

$$w_0(u) = 0.5[1 + \tanh(\frac{u - \theta}{d})]$$

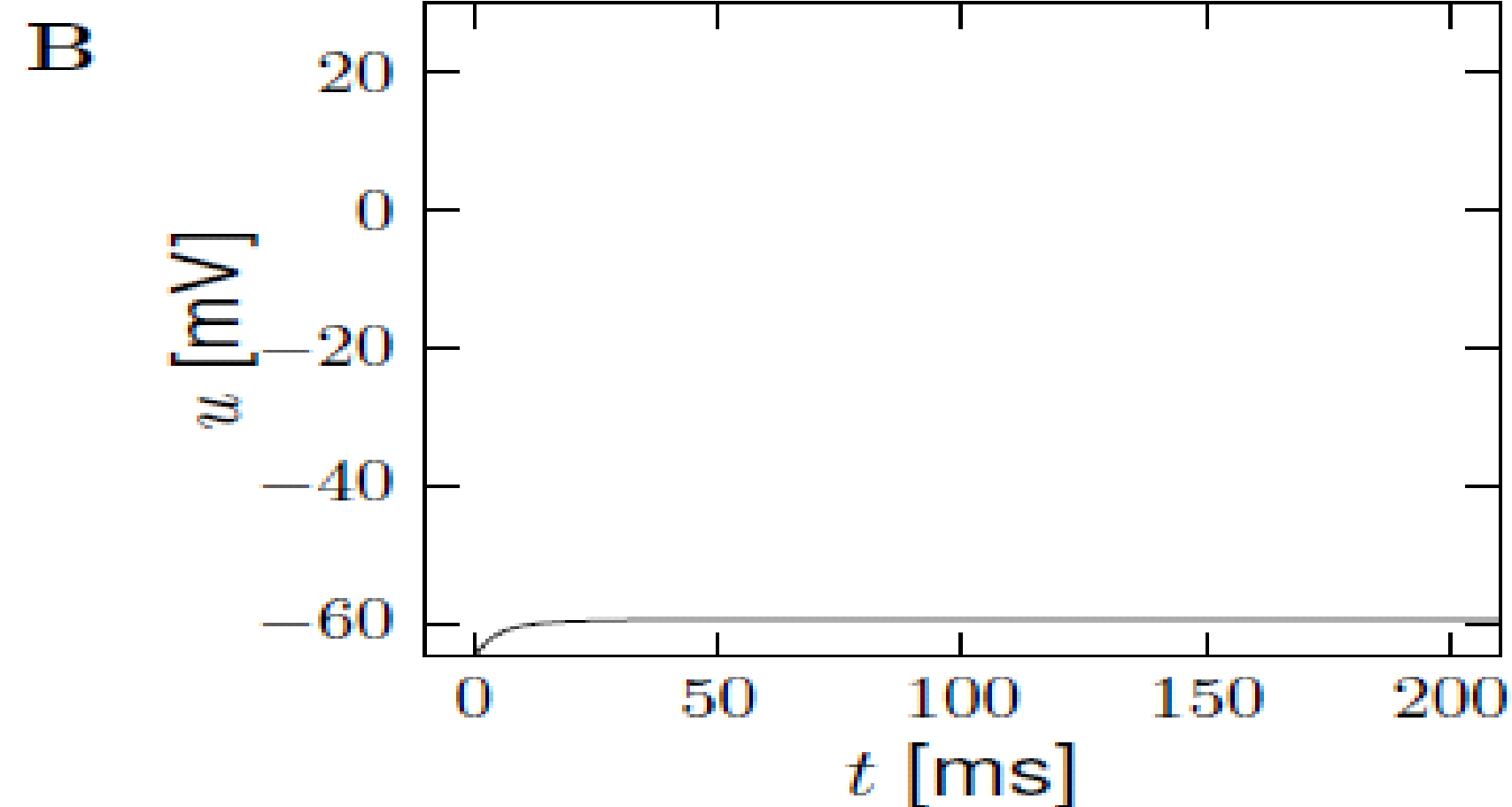
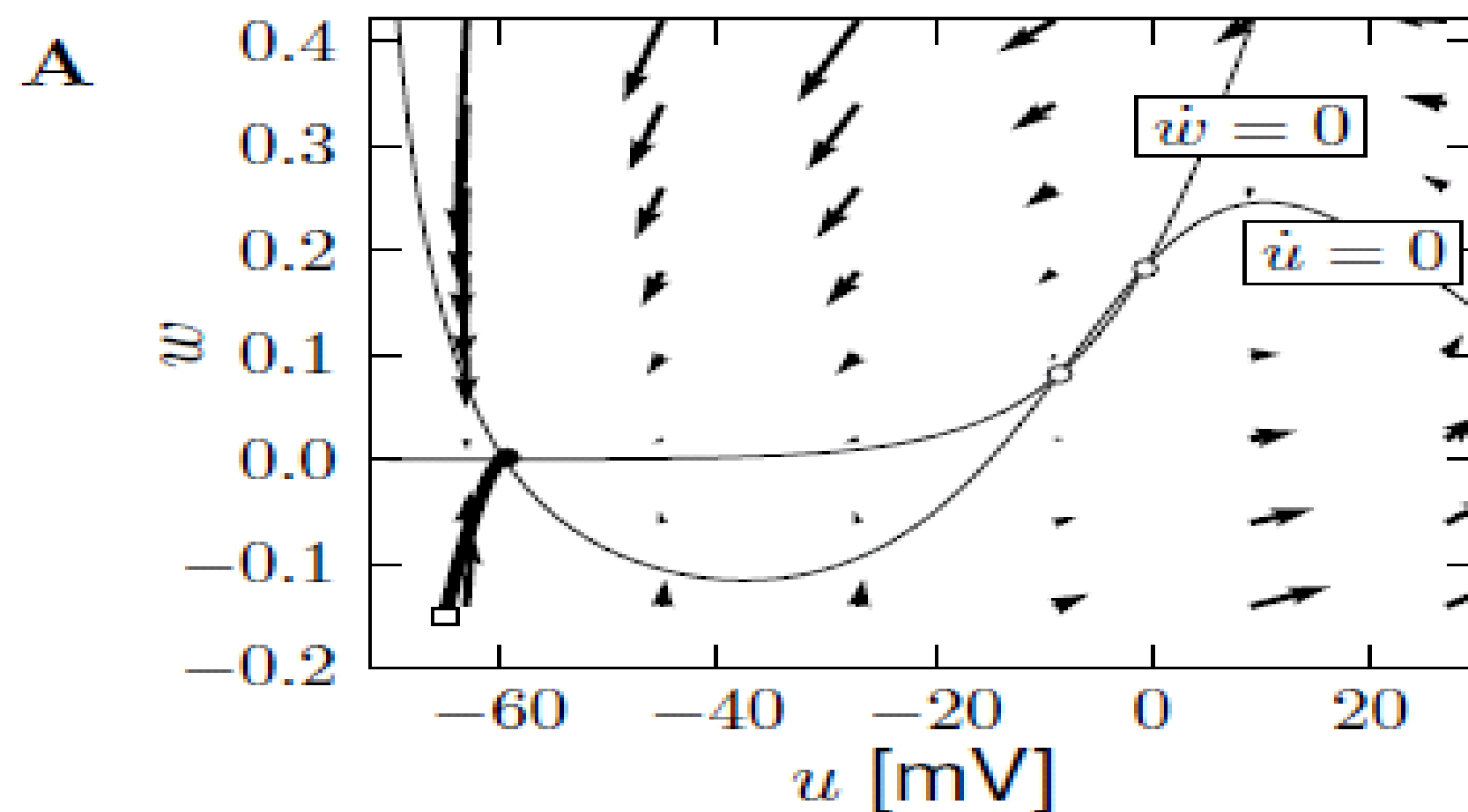


Low-frequency firing

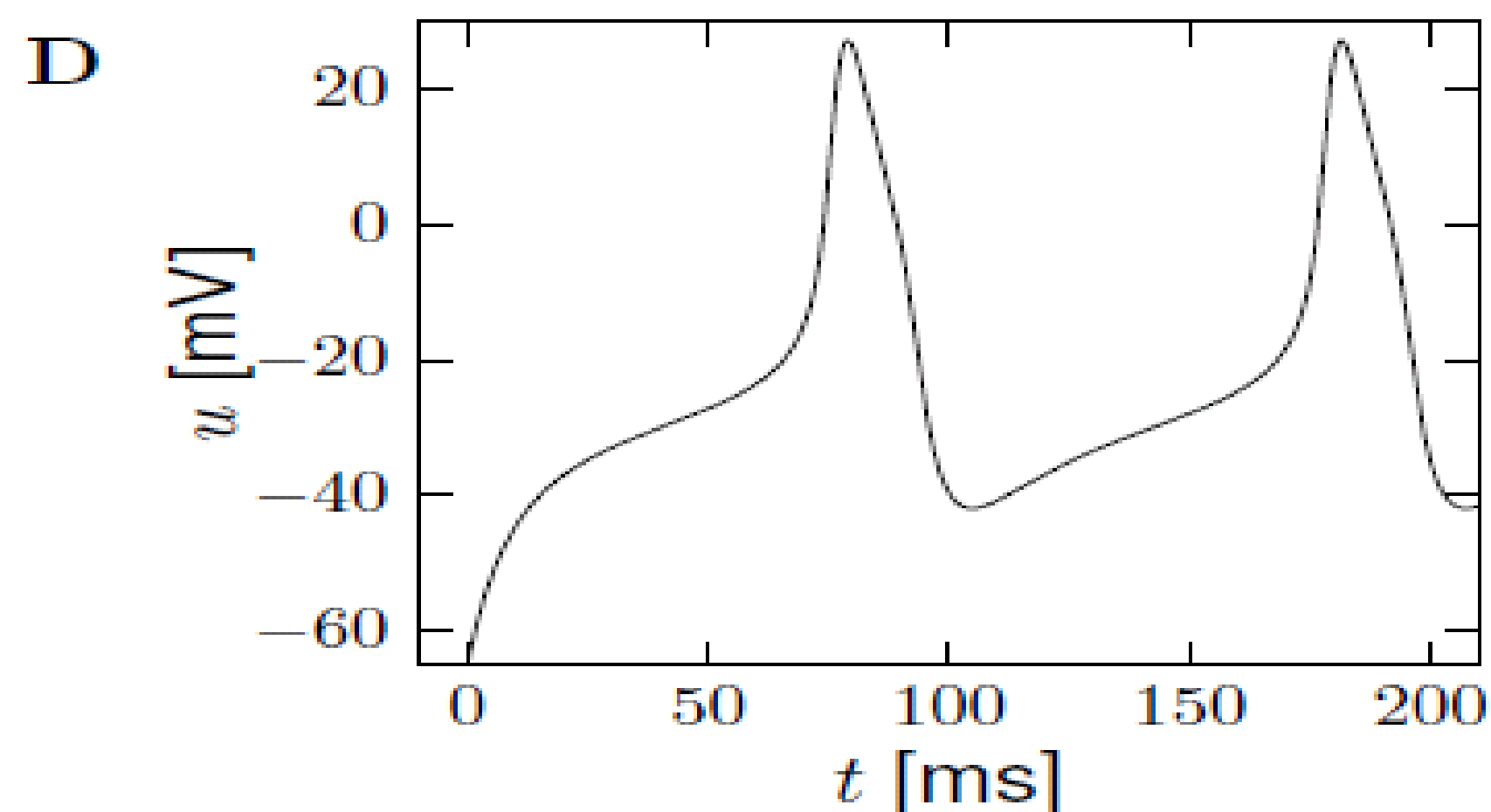
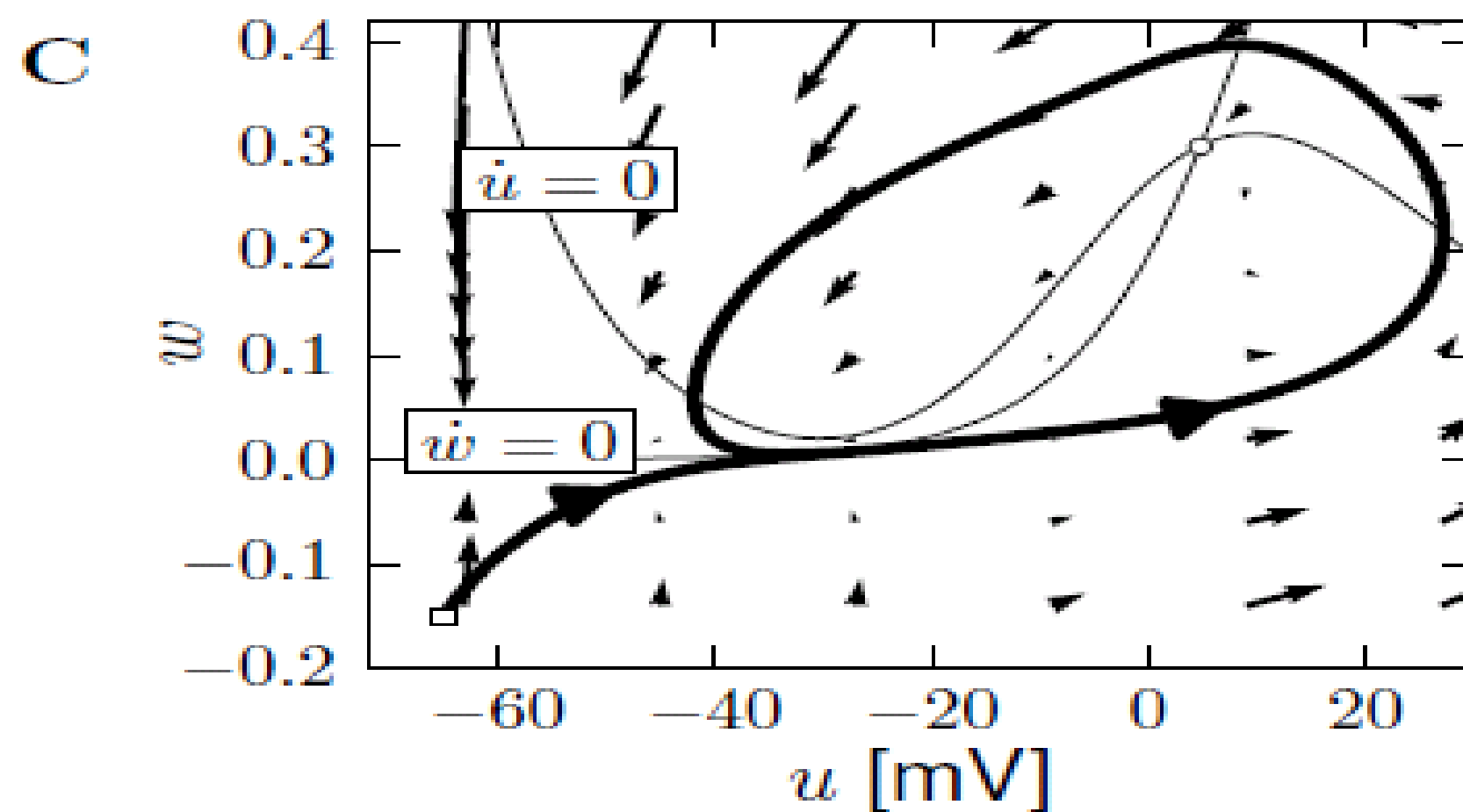


Morris-Lecar, type I Model – constant input

$I=0$



$I > I_c$



Type I and type II models

Response at firing threshold?

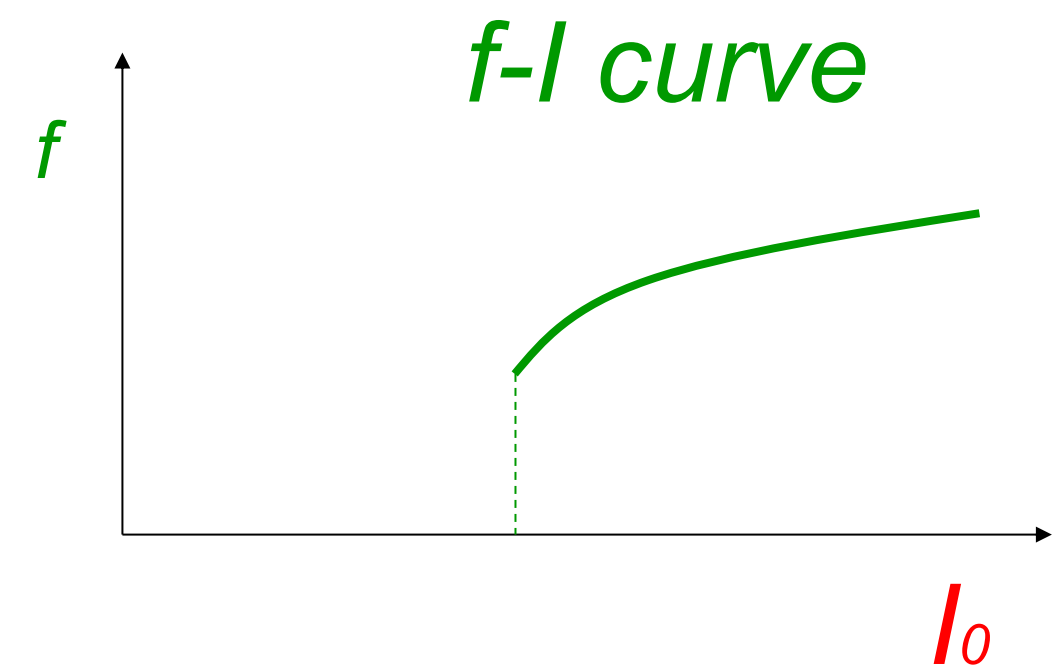
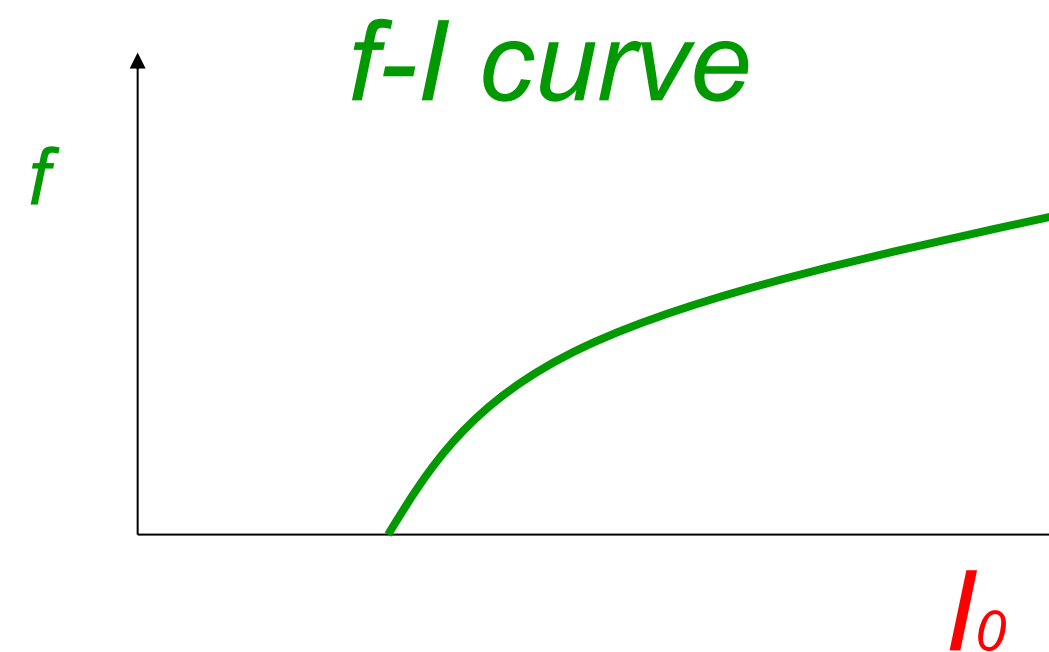
Type I

type II

Saddle-Node
Onto limit cycle

For example:
Subcritical Hopf

ramp input/
constant input



Neuronal Dynamics – Quiz 4.1.

A. 2-dimensional neuron model with (supercritical) saddle-node-onto-limit cycle bifurcation

- ☐ The neuron model is of type II, because there is a jump in the f-I curve
- ☐ The neuron model is of type I, because the f-I curve is continuous
- ☐ The neuron model is of type I, if the limit cycle passes through a regime where the flow is very slow.
- ☐ in the regime below the saddle-node-onto-limit cycle bifurcation, the neuron is at rest or will converge to the resting state.

B. Threshold in a 2-dimensional neuron model with subcritical Hopf bifurcation

- ☐ The neuron model is of type II, because there is a jump in the f-I curve
- ☐ The neuron model is of type I, because the f-I curve is continuous
- ☐ in the regime below the Hopf bifurcation, the neuron is at rest or will necessarily converge to the resting state